THE FRONTIER OF NOVELTY

Machine learning, the unknown, and the architecture of discovery

An artificial-intelligence-produced research survey

Structural limits · movable boundaries · scientific automation

Research updated 24 July 2026

Research disclosure, scope and evidence policy

This book is an artificial-intelligence-produced research survey. It was written by the GPT-5.6 Pro language model under human editorial direction, using web-based research updated through 24 July 2026; GPT denotes generative pre-trained transformer. Artificial intelligence (AI) is used here as the broad term for computational systems that perform tasks associated with perception, reasoning, language, planning or action. Machine learning (ML) refers more narrowly to methods whose behaviour is shaped by data, optimization or experience rather than only by manually specified rules.

The volume is not a peer-reviewed monograph and should not be treated as an authoritative substitute for the primary literature. Its purpose is synthesis: to connect learning theory, uncertainty, distribution shift, causality, continual learning, open-ended search, formal verification, autonomous experimentation and emerging scientific agents around one question. Under what conditions can a computational system recognize, generate and validate something meaningfully new? The answer is necessarily conditional on the evidence, representation, evaluator and scientific institution involved.

The research process prioritized primary peer-reviewed papers, major surveys and official publication records. Preprints and system reports were included when they described frontier work not yet represented in the peer-reviewed literature, but they are identified as such in the prose. News reports and commentary were used mainly to locate debates or corrections and were not treated as equivalent to primary evidence. No claim of exhaustive coverage is made. Rapidly changing results, inaccessible material and errors in automated synthesis remain possible.

Table R.1. Evidence classes used in this survey

Evidence class	How it is interpreted in the book
Peer-reviewed primary research	The strongest source for reported methods and results, while remaining open to replication and correction.
Peer-reviewed review or perspective	Used for field structure, terminology and competing interpretations rather than as direct proof of every claim.
Conference paper or preprint	Treated as current research whose methods and claims may still change; explicitly labelled when central.
Official system report or white paper	Useful for architecture and reported demonstrations, but not equivalent to independent validation.

Evidence class	How it is interpreted in the book
Commentary, editorial or news analysis	Used to frame implications, controversy or institutional context, not as the main support for technical facts.

The table exists because frontier research mixes evidence with very different levels of scrutiny. A spectacular system report can be informative before independent replication, but it should not silently acquire the status of a mature result. Conversely, a theoretical impossibility theorem can be rigorous while applying only under specific assumptions. Throughout the book, evidence is interpreted together with scope. The reader should ask not only whether a result was published, but what claim the method and evaluation actually support.

Terminology has been treated as an editorial requirement rather than an appendix problem. Every abbreviation used in the running text is expanded and defined at first use. Headings avoid unexplained initials. Closely related terms such as anomaly detection, novelty detection, open-set recognition and out-of-distribution detection are distinguished locally before comparison. The glossary at the end is a recap, not a substitute for explanation where the concept first matters.

Figures in the book are conceptual diagrams unless a caption explicitly states otherwise. They do not report experimental datasets. Their purpose is to expose relations that prose can conceal: which components depend on one another, which evidence separates rival claims, and where a system can fail. Tables are limited to two columns so that every comparison answers one clear question. Each table and figure is followed by prose explaining why it exists and what inference should, and should not, be drawn from it.

The survey adopts a verification-first stance. It treats generation as one component of discovery rather than its culmination. It distinguishes structural limits, which follow from absent information, non-identifiability or undecidability, from movable boundaries, which may yield to better data, representations, evaluators, architectures and institutions. This distinction is the book's main defence against both technological fatalism and ungrounded optimism.

This is a map of a moving frontier produced by an artificial-intelligence system. Its strongest use is not as a final authority, but as a structured starting point for criticism, comparison and further research.

Corrections, counterexamples and better sources should be understood as improvements to the survey rather than threats to it. That attitude mirrors the central thesis of the book:

reliable intelligence is not the absence of error but the organization of processes that expose, preserve and correct error.

Contents

A research map of limits, movable boundaries and discovery architectures.

Introduction: From prediction to discovery

Machine learning has become extraordinarily capable at mapping inputs to outputs. It recognizes images, translates language, predicts molecular structures, writes software and controls instruments. These achievements naturally invite a stronger ambition: can computational systems discover what is not yet known? The question is more demanding than asking whether a model can produce an unfamiliar sentence or solve a held-out task. Discovery requires a relation to previous knowledge, a standard of correctness, an account of significance and a process through which errors can be exposed.

The problem begins with the word new. A sample can be new to a training set, statistically unusual, outside learned classes, conceptually different, mechanistically unprecedented or historically unreported. These senses are related but not interchangeable. A system that detects a low-density observation has not thereby discovered a new phenomenon. A generator that produces a candidate absent from its data has not established that the candidate is possible, useful or unknown to humanity. The transition from novelty signal to discovery is an evidential process.

Machine learning does not need to have seen every future answer in order to contribute to discovery. It can learn constraints, representations, search policies and evaluative regularities. It can combine known components in ways not previously instantiated. It can interact with proof assistants, compilers, simulators and laboratories that provide feedback outside the training corpus. The most convincing examples of machine discovery have exactly this structure: a powerful generator is coupled to an evaluator that can reject its mistakes independently.

This constructive claim coexists with real limits. Finite evidence does not determine one unique rule for every unseen case. Unknown distributions cannot be detected universally without assumptions. Causal direction is not always identifiable from observation. A property that leaves no trace in the available measurements cannot be recovered by scale. Historical priority cannot be certified from an incomplete record. Some problems are undecidable or computationally intractable. These are not reasons to stop. They are instructions about what kind of evidence, interaction or claim is required.

A central distinction in the book is between structural limits and movable boundaries. Structural limits arise from information and logic. Movable boundaries arise from current engineering and scientific organization. Calibration under distribution shift is poor but can improve. Continual learning remains brittle but has many architectural levers. Causal representation learning has conditional identifiability results and increasingly realistic testbeds. Automated laboratories can become safer, more auditable and more capable of choosing informative experiments. Scientific agents can be trained and evaluated for evidence use rather than only fluent output.

The frontier is therefore not located at one model scale. It is distributed across representations, memory, search, verification, experimental design and institutions. A model that generates excellent hypotheses but cannot recognize contradictory evidence is not a discovery system. A formal prover with no way to formulate relevant conjectures is equally incomplete. A laboratory that optimizes yield without detecting instrument drift may accelerate error. The scientific machine is the coupled architecture.

The first part develops the epistemology of novelty. It treats novelty as a relation among a candidate, a memory, a representation and a community. It examines induction, identifiability and the technical problem of knowing when a model does not know. The second part studies what happens when the future departs from the training distribution. Distribution shift, shortcut learning, causal ambiguity, continual learning and evaluation are analysed as failures of hidden assumptions.

The third part turns constructive. It develops the generate–search–evaluate–archive pattern and explains why formal, executable and empirical evaluators create unusually fertile regions for machine discovery. Active learning and Bayesian optimization make experiments part of the algorithm. Case studies from mathematical search, protein design, materials and autonomous laboratories show where novelty can be validated and where the claim must remain provisional.

The fourth part examines open-endedness and automated science. Quality-diversity search preserves stepping stones that fixed objectives discard. Self-improving agents modify their own scaffolds under external benchmarks. Scientific-agent systems now perform literature synthesis, hypothesis generation, coding, experiments and manuscript production in bounded domains. Their progress is real, but process evaluations reveal weaknesses in evidence uptake, refutation and belief revision. The final chapters distinguish limits that scale cannot erase from boundaries with plausible research paths and propose a verification-first agenda.

Several themes recur. The first is independence: the evaluator should not simply reproduce the generator's preferences. The second is provenance: a claim should be linked to the data, code, assumptions and decisions that produced it. The third is plurality: rival hypotheses and diverse search lineages should survive long enough to encounter discriminating evidence. The fourth is reversibility: updates, experiments and automated actions should be bounded and correctable. The fifth is institutional design: scientific reliability is produced by communities and tools, not by an isolated model.

These themes alter what progress should look like. A larger benchmark score may be less valuable than a system that abstains under a genuine shift. A faster laboratory may be less valuable than one that detects its own calibration error. A more creative generator may be less valuable than an evaluator that turns one disputed candidate into a verified result. A scientific agent that produces fewer papers but preserves evidence and revises beliefs may contribute more knowledge.

The book does not attempt to settle philosophical questions about consciousness or subjective understanding. It uses an operational notion of understanding: the ability to represent alternatives, transfer a concept, predict interventions, seek disconfirming evidence and revise a model coherently. These behaviours can be tested. They do not establish subjective experience, but they are sufficient for analysing whether a system can participate responsibly in scientific inquiry.

Nor does the book assume that human science is free of the same limitations. Humans also overfit, follow fashion, ignore evidence, use imperfect proxies and misjudge priority. Machine systems make some of these failures more visible because their data and objectives can be inspected. They also make them scalable. The appropriate comparison is not perfect human understanding against mechanical imitation. It is one error-correcting system against another, and the possibility of designing a stronger hybrid.

The important frontier is not a computer that can say something new. It is a system that can locate the uncertainty in a new claim, obtain the evidence that matters, survive an independent test and leave a path that others can reconstruct.

That frontier is already unevenly present. Formal mathematics offers strong verification. Software offers execution. Chemistry and materials offer increasingly autonomous experiments. Biology offers rich perturbation data but difficult translation across levels. Social and institutional questions remain harder because objectives and mechanisms are contested. The chapters that follow treat this unevenness as evidence about architecture rather than as a simple ranking of model intelligence.

PART I

Part I. The Epistemology of the New

Novelty becomes intelligible only after its reference, representation and evaluator are made explicit.

CHAPTER 1

1 Novelty Is a Relation, Not a Substance

1.1 Six different things we call new

The word new seems to name a property of an object, as if novelty were attached to a molecule, theorem, image or event in the same way that mass or colour is attached to it. In practice, novelty is relational. A candidate is new only with respect to a memory, a representation, a comparison rule, a time and a community entitled to make the comparison. The same protein sequence may be new to a model, familiar to a database curator, old in evolutionary history and scientifically novel because nobody had previously recognized its function. Any serious account of machine discovery must therefore begin by replacing the binary question “is it new?” with the more explicit question “new for whom, relative to what record, under which equivalence criterion, at what level of description?”

This relational view immediately separates several problems that are often collapsed into one. A system may determine that it has not stored an exact byte sequence. It may estimate that an observation lies far from the probability mass represented in its training data. It may reject an input because it does not fit any learned class. It may propose a new category that compresses a cluster of observations. It may infer a mechanism that explains regularities not captured by existing theory. It may also claim historical priority. These acts require progressively more evidence. Exact memory comparison is mostly a retrieval problem. Scientific novelty is a social and epistemic judgment whose evidence includes prior literature, semantic equivalence, experimental validity and the significance of the difference.

Table 1.1. Forms of novelty and the evidence each one requires

Form of novelty	What would count as evidence
Unseen instance	A traceable comparison against the system’s accessible memory or database.
Statistical anomaly	A deviation measured in a declared representation relative to a reference distribution.
Unknown class	Evidence that no known class provides an adequate fit under an explicit rejection rule.
New concept	A stable abstraction that improves explanation, compression, prediction or intervention across cases.

Form of novelty	What would count as evidence
New mechanism	Discriminating observations or interventions that favour one causal account over alternatives.
Historical priority	A defensible search for equivalent prior work across literature, patents, archives and communities.

The table exists to prevent a recurrent category error. A high anomaly score can support the claim that an input is unusual in a chosen feature space, but it cannot by itself support the claim that science has encountered a new entity. Likewise, absence from a training corpus does not imply absence from the world's literature. By placing each sense of novelty beside the evidence appropriate to it, the table turns an attractive word into a set of testable claims. The practical consequence is that systems should report the level of novelty they can actually defend rather than silently upgrading a local mismatch into a universal discovery.

The role of representation is especially important. A detector that represents animals by colour and silhouette may judge a thermal image of a familiar cat to be radically new while failing to distinguish a genuinely new species whose visible outline resembles a known one. A language model may regard two proofs as different strings while a mathematician recognizes them as the same argument under a change of notation. A materials system may identify a composition as novel even though its crystal structure and function duplicate earlier work. Novelty is therefore never independent of the features, invariances and equivalence relations built into the comparison.

Equivalence is not a technical detail that can be postponed. In mathematics, one may compare formulas syntactically, extensionally or up to isomorphism. In chemistry, novelty may refer to a molecular graph, a stereoisomer, a synthesis route or a measured property. In scientific explanation, two verbal accounts may be different descriptions of the same mechanism, or nearly identical language may conceal incompatible causal commitments. A discovery system requires domain-specific ways to decide when two candidates are "the same enough." Without them, an archive fills with superficial variants and novelty becomes an artefact of notation.

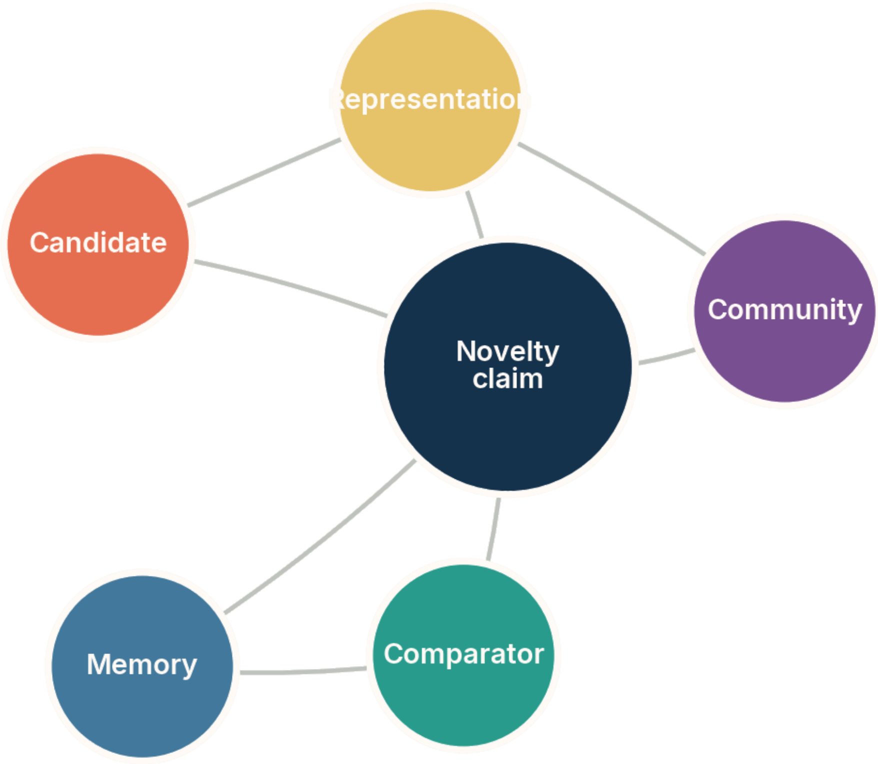


Figure 1.1. A novelty claim emerges from a relation among a candidate, a memory, a representation, a comparator and a community. Conceptual diagram; not an empirical result.

The figure matters because it locates novelty in a network of dependencies rather than inside the candidate alone. Removing any node changes the claim. Without memory there is no reference set; without representation there is no declared basis of comparison; without a comparator there is no operational criterion; without a community there is no stable convention for significance or priority. The drawing also explains why improved models do not automatically solve novelty. A more capable generator may produce better candidates while the comparison record, equivalence rule or institutional judgment remains weak.

For machine learning, this distinction suggests a layered reporting discipline. A system can safely state that a sample is absent from a specified database, that its score falls in a low-density region of a representation, or that a classifier abstained because the example did not match known classes. It should reserve stronger language such as “new concept,” “new mechanism” or “first discovery” for cases in which additional evidence has been collected. Such discipline is not merely rhetorical. It determines which errors can be audited and which downstream decisions should be automated.

The deepest point is that novelty is not identical to surprise. A predictable event can be historically unprecedented, and a surprising event can be an instrument failure. Nor is novelty identical to value. Random noise is maximally difficult to predict but scientifically empty. The frontier of machine discovery is therefore not a machine that maximizes unfamiliarity. It is a system that can distinguish unfamiliarity from error, connect differences to explanatory structure, and seek the additional observations needed to turn a provisional novelty signal into reliable knowledge.

1.2 Discovery, invention, surprise and significance

Scientific discourse often treats discovery as the moment when an answer is generated. That image is misleading. A candidate result becomes a discovery only after several thresholds have been crossed: it must be sufficiently distinct from what is already known, correct under the relevant standards, connected to a question that matters, and communicable in a form that others can inspect and reuse. Invention emphasizes construction; discovery emphasizes a warranted claim about a structure, regularity or possibility. Machine learning can participate in either, but the evidential route differs.

Consider an algorithm that performs a computation faster than the best method in a benchmark. The candidate may be invented through search, but its status as a discovery depends on executable verification, comparison against previous algorithms and a precise account of the cost model. Consider a generated molecule with a novel graph. It is not yet a drug, a mechanism or even a synthesizable compound. The candidate must survive chemical constraints, synthesis, measurement and often biological validation. In each case, generation expands the space of possibilities; verification contracts that space into claims that can be trusted.

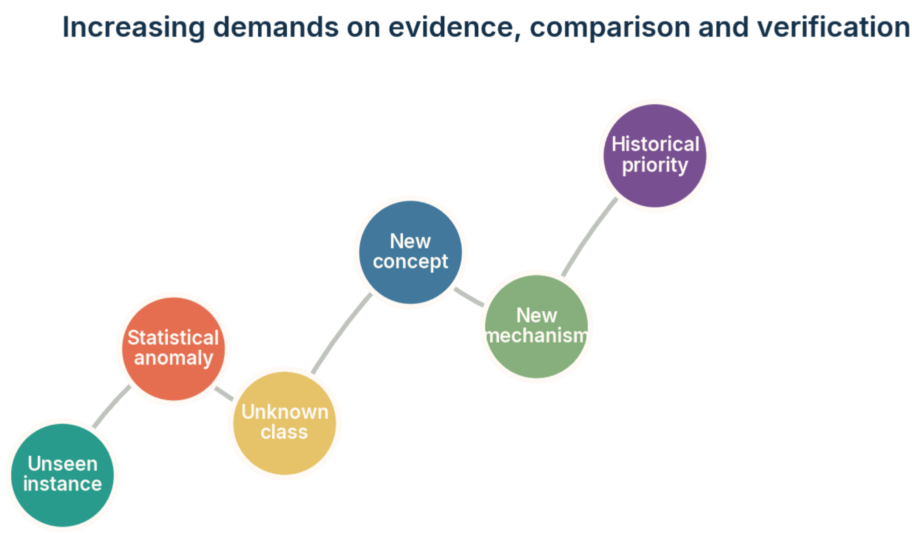


Figure 1.2. Different novelty claims demand progressively stronger evidence and broader comparison. Conceptual diagram; not an empirical result.

The figure matters because it shows that novelty is not a single threshold. A system may be excellent at the left side of the spectrum and unreliable at the right. Memory systems and nearest-neighbour methods can often detect exact or approximate repetition. Statistical detectors can rank atypicality. Open-world classifiers can sometimes reject unfamiliar categories. New concepts and mechanisms require explanatory work, while historical priority requires an external record that no model can make complete. The increasing evidential burden is the reason that discovery architectures need multiple evaluators rather than one universal novelty score.

Surprise occupies an ambiguous position in this spectrum. In Bayesian terms, an observation is surprising when it has low probability under a current model. That is useful because surprise can identify where a model is inadequate. Yet a low-probability observation can arise from sensor corruption, sampling bias, adversarial manipulation or a rare but understood event. Scientific surprise becomes productive only when a system asks why the observation was improbable and identifies tests that distinguish model failure from data failure. The value lies not in the magnitude of surprise but in the quality of the revision it provokes.

Significance introduces another distinction. A candidate can be novel without altering what anyone can explain or do. Search systems are particularly vulnerable to this problem because large generative spaces contain endless variants that are different under a weak metric. The concept of minimal description length provides one useful intuition: a new hypothesis is valuable when it compresses observations or replaces a collection of exceptions with a simpler reusable structure. Predictive improvement, causal control, proof simplification, robustness across environments and the opening of a new design region are other possible standards. No single standard applies everywhere, but every discovery claim needs one.

Table 1.2. Four thresholds between a generated candidate and a scientific contribution

Threshold	Question that must be answered
Distinctness	Is the candidate meaningfully different under a domain-appropriate equivalence rule?
Validity	Does it satisfy formal, executable, empirical or observational constraints?
Significance	Does it improve explanation, capability, compression, prediction or intervention?
Priority and provenance	Can the lineage of the claim, data, code and comparison with prior work be reconstructed?

The table exists because contemporary artificial-intelligence systems often leap from distinctness directly to contribution. A language model produces an unusual hypothesis, another model rates it as plausible, and the result is described as novel. The missing thresholds are exactly where hallucination, duplication and triviality enter. Validity must be grounded in an evaluator that is at least partly independent of the generator. Significance must be defined relative to a scientific objective rather than stylistic impressiveness. Priority requires a documented search and remains provisional even after careful review.

These thresholds also clarify why automated discovery is uneven across domains. In program synthesis, a test suite or formal specification can reject many incorrect candidates cheaply. In theorem proving, a proof assistant can check a derivation line by line. In materials science, evaluation may require expensive synthesis and characterization. In social science, the relevant mechanism may depend on institutions and unobserved variables, making validation slower and more ambiguous. The same generative model can therefore appear brilliant in one domain and speculative in another because evaluator strength, not linguistic fluency, is the limiting resource.

The distinction between discovery and invention further suggests complementary roles for humans and machines. Machines can search enormous combinatorial spaces, preserve diverse candidates and execute repetitive tests. Humans remain unusually effective at redefining questions, recognizing changes in significance, negotiating standards of evidence and noticing when an apparently technical result alters the conceptual map of a field. This division is not fixed, but it reminds us that automating candidate production is different from automating the social process by which a community decides what deserves to count as knowledge.

> *A machine does not discover merely by saying something that was not in its training data. It discovers when a traceable process turns a candidate into a claim that survives comparison, criticism and contact with the world.*

The constructive agenda follows directly. Research should improve novelty databases and semantic equivalence checks, design evaluators that cannot be easily gamed, represent uncertainty about priority, and build systems that actively request missing evidence. It should also measure the diversity and significance of discoveries rather than only the volume of generated output. The relevant frontier is not unrestricted originality. It is disciplined expansion of the space in which reliable claims can be made.

CHAPTER 2

2 Induction, Identifiability and Impossibility

2.1 Finite evidence, inductive bias and the price of generalization

Machine learning begins with an asymmetry: the evidence is finite, but the claims usually concern cases not yet observed. A training set constrains a model, yet in any sufficiently expressive hypothesis class many rules can agree on every training example and disagree elsewhere. Generalization is therefore never obtained from data alone. It depends on inductive bias, the collection of preferences and structural assumptions that make some continuations of the evidence more likely than others. Architecture, optimization, regularization, data augmentation, pretraining and the choice of representation all contribute to that bias.

The need for bias is not a defect introduced by neural networks. It is a condition of induction. Tom Mitchell's early formulation made the point directly: a learner that generalizes must prefer some hypotheses over others. No Free Lunch theorems for optimization make a related claim. Averaged uniformly across all possible objective functions, no search method has a universal advantage (Wolpert and Macready, 1997). Practical success is possible because real problems are not uniformly arbitrary. Images, language, physical systems and scientific data contain regularities, symmetries, locality and compositional structure. The open question is which biases capture those structures without becoming brittle when the world changes.

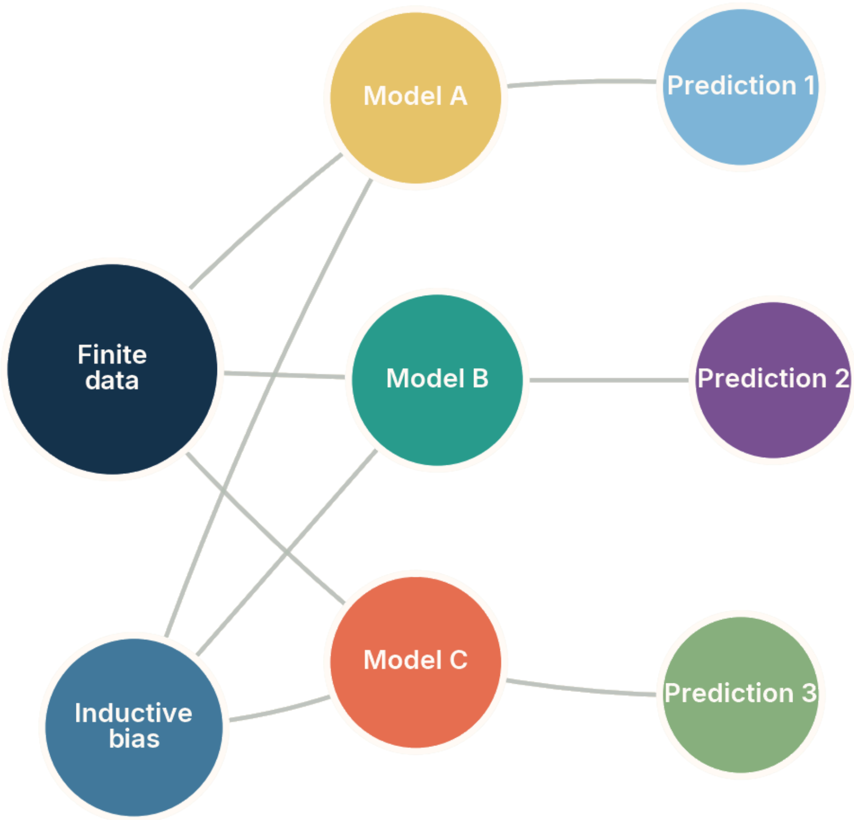


Figure 2.1. The same finite data can support several models that make incompatible predictions beyond the observed region. Conceptual diagram; not an empirical result.

The figure matters because it visualizes underdetermination without treating it as a rare pathology. Model A, Model B and Model C can fit the same evidence. Their divergence appears only on a new case. The inductive bias node represents the often-hidden reason that one model is selected: perhaps it is simpler, smoother, more causal, more compatible with a pretrained representation or easier to optimize. When a system presents a prediction without exposing these dependencies, confidence can be mistaken for logical necessity.

Scaling changes this picture but does not abolish it. Larger models trained on broader data can acquire useful priors that resolve many ambiguities encountered in ordinary tasks. They may interpolate across regions that were previously sparse and transfer regularities between domains. Yet broader priors can also make assumptions harder to see. A foundation model carries the statistical history of its corpus, annotation practices, tool use and optimization objective. The model may generalize impressively because the new task resembles structures already represented, while failing abruptly where those structures cease to apply.

A useful distinction is between interpolation in a rich representation and extrapolation under a changed mechanism. What looks like radical novelty to a human observer may be close to known examples in the model's latent space. Conversely, a visually minor change may alter the causal process that generated the data. Generalization claims should therefore specify the axis along which novelty occurs. New combinations, new parameter ranges, new causal regimes, new sensors and new goals pose different challenges. A single test score cannot summarize them.

Table 2.1. Common inductive biases and the worlds in which they help

Bias or assumption	When it is useful and when it can fail
Smoothness	Useful when nearby inputs have nearby outputs; brittle near discontinuities and regime changes.
Locality and translation symmetry	Powerful for spatial data; misleading when global context or absolute position is decisive.
Simplicity or compression	Favors reusable explanations; can prefer elegant but false models when reality is heterogeneous.
Compositionality	Supports recombination of known parts; fails when interactions create qualitatively new behaviour.
Causal invariance	Supports transfer across environments when mechanisms remain stable; fails when the mechanism itself changes.
Pretrained semantic structure	Provides broad priors; can import historical bias, omissions and spurious associations.

The table exists to make explicit that every route to generalization is conditional. A bias is neither simply good nor bad. It is a bet on the structure of the world. Machine-learning research often compares methods on average performance while leaving the bets implicit. A more scientific evaluation identifies which structural assumptions each method embodies and deliberately tests environments where those assumptions hold, weaken or fail. This converts surprising failure from an embarrassment into evidence about the learner's effective theory.

The frontier lies in learning biases that are both powerful and revisable. Meta-learning attempts to infer learning strategies from families of tasks. Causal representation learning searches for variables and mechanisms that remain stable across interventions. Equivariant architectures encode known symmetries, while neural program induction and modular systems aim to support compositional recombination. None eliminates induction. Their promise is to move assumptions from accidental correlations toward structures that are more stable, interpretable or testable.

Another frontier is selective prediction. When several hypotheses remain compatible with the evidence, a system should preserve the ambiguity rather than collapse it prematurely. Ensembles, Bayesian approximations, conformal methods and explicit version spaces offer different ways to represent alternatives. An active system can then select observations that separate them. This changes the objective from "guess the unseen label" to "identify which missing evidence would most reduce underdetermination." The latter is closer to the logic of scientific inquiry.

The central limit can therefore be stated positively. Finite data do not determine a unique future, but they can support a disciplined set of possibilities. Progress comes from making the assumptions visible, testing them across environments, maintaining rival explanations and choosing experiments that force those explanations to disagree. Machine learning becomes a discovery method not when it escapes induction, but when it manages induction as an explicit, revisable commitment.

2.2 Identifiability, observability and limits that more data do not remove

Some learning problems are difficult because available algorithms are immature. Others are impossible under the information provided. The concept that separates these cases is identifiability. A target is identifiable when distinct underlying explanations produce distinguishable observations under the stated assumptions. If two possible worlds generate exactly the same accessible data but require different answers, no learner can always choose correctly. More computation cannot recover a distinction that leaves no trace in the input.

Identifiability appears throughout machine learning. A mixture model may admit multiple parameterizations with the same likelihood. Latent factors can be rotated or permuted without changing observations. Causal graphs can be observationally equivalent. A classifier cannot distinguish categories that are identical in the measured features. Unsupervised disentanglement is not identifiable without assumptions that connect latent variables to the data-generating process (Locatello et al., 2019). These are not signs that models are insufficiently intelligent. They are statements about information.

Observability is the engineering counterpart. A relevant property may exist but remain inaccessible to the sensors, sampling procedure or record. A medical model cannot infer a physiological process that affects neither the measurements nor correlated variables. A fraud detector cannot reliably identify a novel strategy engineered to mimic all observed statistics of legitimate behaviour. A scientific agent cannot establish historical priority from a corpus that omits unpublished, inaccessible or differently described work. In each case, the missing channel matters more than model size.

Table 2.2. Four classes of limits that are often mistaken for one another

Class of limit	Diagnostic question
Information limit	Does the available evidence contain any signal that distinguishes the alternatives?
Identification limit	Given the assumptions, does one underlying explanation follow uniquely from the evidence?
Computational limit	Is the answer defined but intractable to obtain with feasible resources?
Evaluation limit	Can correctness or progress be measured without relying on the same fallible system that generated the claim?

The table exists because the remedy depends on the diagnosis. An information limit requires new sensors, interventions, records or labels. An identification limit requires stronger assumptions or a redesigned experiment. A computational limit may be attacked with approximation, decomposition, hardware or a restricted problem class. An evaluation limit requires a stronger verifier or a different institutional process. Treating all four as "model capability" encourages wasteful scaling and obscures the actual bottleneck.

Computability adds a separate boundary. Some formally defined problems are undecidable in general; others are decidable but require resources that grow too rapidly. Machine learning can provide useful heuristics on distributions of practical instances, but it does not repeal these results. A theorem-proving model may solve many statements while no general procedure can decide every mathematical proposition in a sufficiently expressive formal system. A program-analysis agent may detect numerous bugs while the halting problem remains undecidable. The practical frontier is distributional competence under explicit coverage, not universal guarantees.

Logical limits also constrain self-verification. A powerful system cannot in general establish every true statement about its own behaviour from within a fixed formal framework. In practice the more immediate issue is mundane: a model used both to generate and judge an answer shares blind spots with itself. Independent tools, formal checkers, alternative models, human review and physical replication reduce correlated error. They do not produce absolute certainty, but they create epistemic separation between proposal and acceptance.

The distinction between structural and movable limits should guide research investment. Structural limits include absent information, observational equivalence under the allowed data, undecidability and the impossibility of certifying global historical novelty from an incomplete record. Movable boundaries include poor sensor design, weak representations, inefficient search, limited calibration, brittle adaptation and inadequate evaluation. The latter can advance dramatically. The former can often be circumvented by changing the problem, adding interventions or narrowing the claim, but not by pretending the constraint does not exist.

When two worlds W_1 and W_2 induce the same distribution over every observation available to the learner, yet require different answers to a target question, no decision rule based only on those observations can be correct in both worlds.

This statement is a compact test for overclaiming. It asks us to construct rival worlds, not merely rival models. If the evidence would look the same in both, then confidence must come from an assumption that rules one world out. The assumption may be reasonable, such as temporal order, sparsity, smoothness or invariance, but it should be named and tested. Scientific progress often consists of designing an intervention that makes the worlds observationally different.

The constructive consequence is an architecture of epistemic escalation. A system begins with passive observations and a set of compatible explanations. It searches for missing variables, proposes measurements or interventions, estimates the cost and risk of obtaining them, and updates the set of explanations after evidence arrives. In this architecture, admitting non-identifiability is not failure. It is the signal that triggers better experimental design.

CHAPTER 3

3 Knowing When the Model Does Not Know

3.1 Anomaly detection, novelty detection, open-set recognition and out-of-distribution detection

The phrase "detect the unknown" covers several technical tasks with different assumptions. Anomaly detection searches for rare or irregular observations, often within a nominally single distribution. Novelty detection learns a description of normal data and flags future departures. Open-set recognition asks a classifier to identify known classes while rejecting examples that belong to classes absent during training. Out-of-distribution (OOD) detection asks whether a test input comes from the training distribution or from a different one. The acronym OOD will be used only in this technical sense throughout the book.

These tasks overlap but are not interchangeable. A rare example can be in distribution. An example from a different distribution can look ordinary. A new class can be embedded inside the same region of feature space as known classes. An anomaly detector may be trained without labels, while open-set recognition depends on known class boundaries. The operational question, the reference distribution and the costs of false acceptance and false rejection must therefore be specified before a score has meaning.

Table 3.1. Related methods for handling unfamiliar inputs

Technical setting	Core question and hidden dependency
Anomaly detection	Is this observation atypical relative to a nominal population? It depends on how rarity is represented.
Novelty detection	Does a future observation depart from a learned description of normality? It depends on coverage of normal variation.
Open-set recognition	Does the input belong to a known class or should it be rejected? It depends on separability of unknown classes.
Out-of-distribution detection	Does the input arise from the reference distribution? It depends on which alternative distributions matter.
Change-point detection	Has the data-generating process changed over time? It depends on temporal assumptions and detection delay.

Technical setting	Core question and hidden dependency
Selective prediction	Should the model answer or abstain? It depends on a utility or risk threshold, not novelty alone.

The table exists to stop methods from inheriting claims they were not designed to support. An anomaly score is not automatically a probability that an object belongs to an unknown semantic category. A distribution detector does not necessarily explain the nature of the shift. Selective prediction may abstain on difficult familiar cases and answer confidently on unfamiliar ones. Treating these methods as variants of a universal "unknown detector" hides the assumptions under which each can work.

A common design learns a score $s(x)$ and rejects inputs whose score crosses a threshold. The score may use softmax confidence, energy, distance to class prototypes, density estimates, reconstruction error, gradients, ensembles or representations learned from auxiliary data. Evaluation often uses the area under the receiver operating characteristic curve (AUROC), which measures ranking across thresholds, alongside threshold-specific error rates. These metrics are useful only relative to chosen in-distribution and out-of-distribution datasets. A detector can rank one family of shifts well and fail on another.

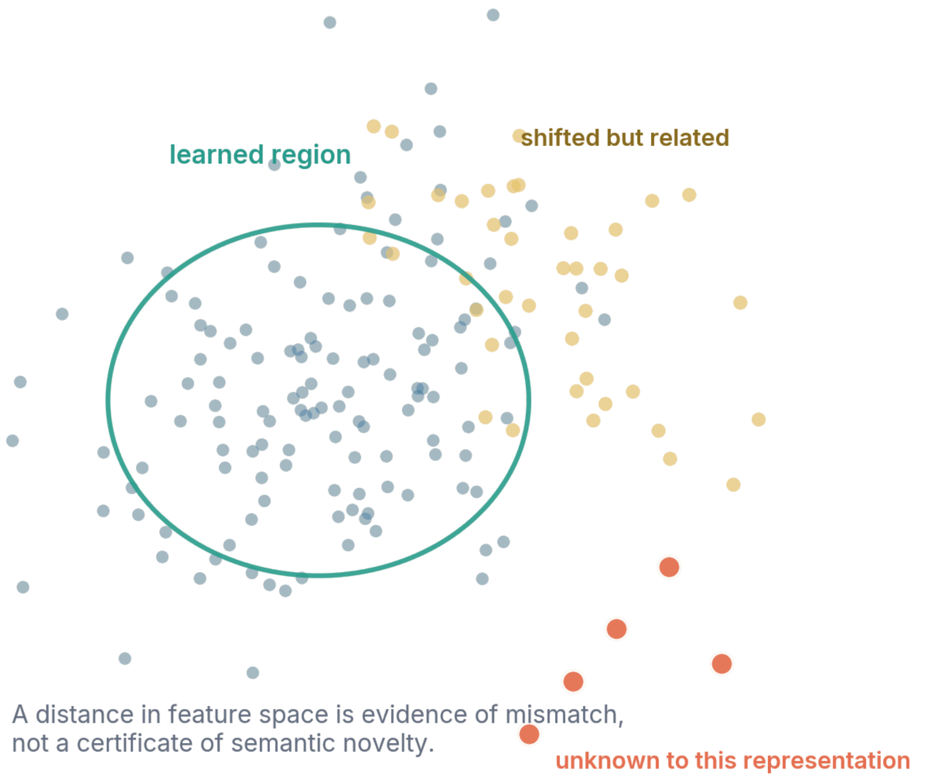


Figure 3.1. Out-of-distribution detection measures mismatch in a representation; it does not certify semantic or historical novelty. Conceptual diagram; not an empirical result.

The figure matters because the green boundary is not a border of the known world. It is a boundary induced by features, training data and a scoring rule. Yellow points may be shifted but still governed by familiar mechanisms. Red points may be sensor failures rather than discoveries. A genuinely new phenomenon can also appear inside the learned region if the representation discards the feature that makes it new. The drawing therefore turns the detector's output into the correct statement: "this observation mismatches my reference in this representation."

The theoretical limit is sharp. Fang and colleagues show that out-of-distribution detection is not universally learnable without restrictions on the distributions and hypothesis classes involved (Fang et al., 2024). Intuitively, if the unknown distribution can be arbitrary, it can imitate the known distribution wherever the learner observes data and differ elsewhere. No detector can guarantee success against every possible alternative. Practical methods work by exploiting structure: semantic separation, auxiliary outliers, pretrained features, density regularities or assumptions about how shifts occur.

This result does not make detection futile. It clarifies the research problem. A deployable system should define an “OOD threat model” analogous to a safety case: which changes are expected, which sensors may fail, which unknown classes are costly, how thresholds will be chosen, and what happens after rejection. Benchmarks should include near shifts that preserve superficial appearance, far shifts, semantic novelties, nuisance changes and adversarially selected cases. Performance should be reported separately rather than averaged into a reassuring number.

Promising frontier work combines representation learning with active escalation. A detector can group rejected cases, retrieve related evidence, request labels, trigger additional sensors or route a case to a specialist. Human feedback can improve coverage of operationally relevant unknowns, while online methods can update thresholds as environments change. The goal is not to label every surprise correctly at first contact. It is to keep mismatch from becoming silent confident error and to convert recurring unfamiliarity into a new, audited region of competence.

The correct ambition is thus bounded but powerful. Machine learning can learn the geometry of a known region, estimate where that geometry becomes unreliable and construct procedures for abstention and investigation. It cannot learn the shape of every possible unknown in advance. The research frontier lies in making the reference explicit, broadening the representations, learning from rejected cases and linking detection to evidence-gathering actions.

3.2 Uncertainty, calibration, abstention and conformal prediction

A model can be wrong for many reasons, and its numerical confidence is not automatically a measure of those reasons. Predictive uncertainty may reflect ambiguity in the observed case, insufficient data about the model, distribution shift, label noise or unresolved alternatives. Deep classifiers are often miscalibrated: among predictions assigned a nominal probability of ninety per cent, the observed frequency of correctness need not be ninety per cent (Guo et al., 2017). Under dataset shift, even methods that appear calibrated on held-out data can deteriorate substantially (Ovadia et al., 2019).

Calibration is relational in the same way novelty is relational. A probability may be calibrated over a population and time period while being unreliable for a subgroup or a changed environment. Average calibration can coexist with dangerous local overconfidence. Language models add a further complication because verbal expressions such as “likely” or “I am uncertain” are generated tokens, not privileged access to an internal epistemic state. Research on semantic entropy shows that variation across meaning-equivalent generations can help identify some hallucinations, but it remains a task-specific signal rather than a universal introspective faculty (Farquhar et al., 2024).

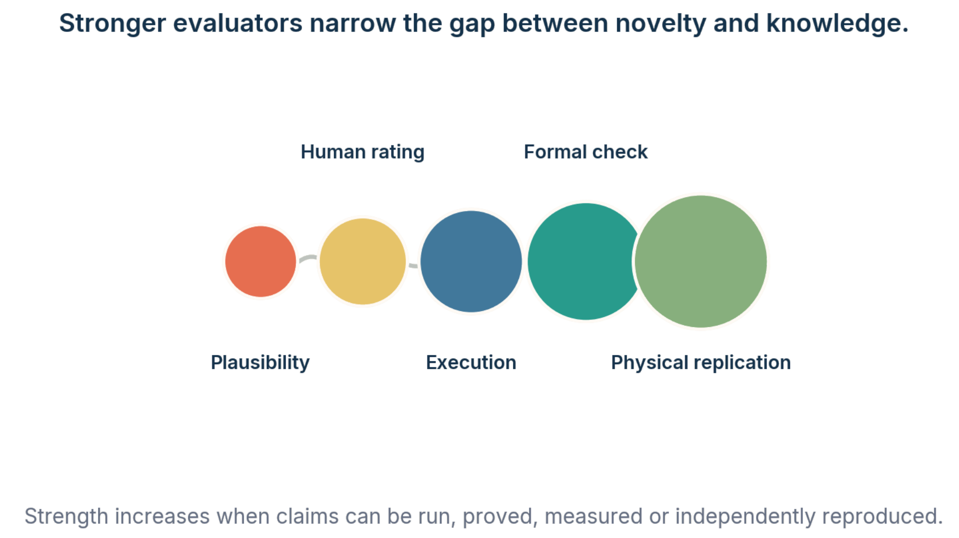


Figure 3.2. Confidence becomes more useful as it is coupled to increasingly independent forms of verification. Conceptual diagram; not an empirical result.

The figure matters because uncertainty estimates are not substitutes for verification. A plausibility score may help prioritize candidates, but execution can reveal errors that plausibility misses. Formal checking can certify properties within a specification. Physical replication can test whether the specification captures the world. Stronger evaluators do not merely increase confidence; they change the kind of claim that can be defended. The practical design principle is to route high-stakes uncertainty toward evaluators with greater independence from the generator.

Selective prediction gives the model a reject option. Instead of forcing an answer for every input, the system abstains when estimated risk exceeds a threshold. The threshold should be chosen by decision cost: a medical screening system, a content filter and a recommendation engine have different tolerances for false acceptance and false rejection. Abstention is meaningful only when a reliable fallback exists. A system that refuses difficult cases but sends them to an equally fallible model has rearranged uncertainty rather than managed it.

Conformal prediction offers a different approach. Under exchangeability assumptions, it converts model scores into prediction sets with finite-sample coverage guarantees. For classification, the output may be a set of labels rather than one label; for regression, an interval rather than a point. The guarantee is marginal over future examples drawn under the stated conditions, not a promise that every subgroup or individual interval is correct. Online conformal methods extend the idea to some forms of distribution shift, but they trade stronger assumptions, delayed feedback or adaptive coverage for the classical guarantee (Angelopoulos and Bates, 2023; Gibbs and Candès, 2024).

Table 3.2. What common uncertainty tools can and cannot guarantee

Method	Useful guarantee and principal limitation
Calibrated probabilities	Align stated confidence with empirical frequency on a reference population; calibration can break locally or after shift.
Deep ensembles	Capture disagreement among trained models; members may share data, architecture and blind spots.
Bayesian approximations	Represent parameter uncertainty under a model; results inherit the model class and prior assumptions.
Conformal prediction	Provides coverage under exchangeability or adapted online assumptions; sets may become wide or uninformative.
Abstention	Limits automated decisions when estimated risk is high; value depends on threshold design and fallback quality.
External verification	Tests a claim through execution, formal proof or measurement; may be costly, incomplete or based on a flawed specification.

The table exists to replace the question "which uncertainty method is best?" with "which uncertainty claim is required?" Methods describe different layers of ignorance. Ensembles estimate sensitivity to training variation, conformal methods control coverage, abstention manages decisions and external verification tests the object itself. Combining them is often more defensible than selecting one scalar score. A discovery system might use an ensemble to identify disputed hypotheses, conformal sets to preserve alternatives, and experiments to resolve them.

A frontier direction is uncertainty decomposition tied to action. Instead of one confidence number, a system can estimate whether uncertainty is likely to be reduced by retrieving literature, collecting another measurement, changing the model class or asking a human. This is a value-of-information problem. The output is not simply "I do not know," but "the decision is sensitive to this missing variable, and this observation is expected to reduce that sensitivity." Such systems would turn metacognition into an operational research policy.

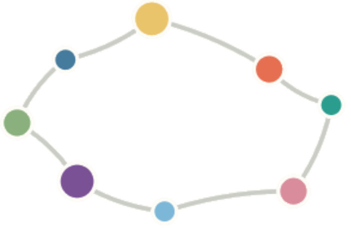
Another direction is calibration under evolving conditions. Static test sets are insufficient when deployed systems change their users and data. Living calibration monitors residuals, subgroup performance, abstention rates and the composition of rejected cases. It can detect when confidence scores stop meaning what they meant at release. This requires provenance and delayed ground truth, but it is one of the most realistic ways to prevent uncertainty estimates from becoming ceremonial.

The broader lesson is that knowing what one does not know is not a single learned capability. It is an architecture that connects scores, assumptions, decision costs, fallback procedures and independent evidence. Machine learning can make major progress here, but no internal confidence signal can certify that all unknown unknowns have been captured. Reliable systems remain open to correction from outside themselves.

PART II

Part II. Beyond the Training Distribution

A model leaves the training distribution long before an organization admits that its assumptions have changed.



CHAPTER 4

4 When the World Changes

4.1 Distribution shift, shortcut learning and underspecification

A model is usually trained and evaluated under an approximation that the future resembles the recorded past. Distribution shift names the many ways in which that approximation can fail. The frequency of inputs can change, the relation between inputs and labels can change, the population can change, the sensor can change, the decision rule can alter behaviour, and the model's own deployment can reshape the environment. These are not cosmetic variants of one problem. Each shift changes which parts of the learned representation remain relevant.

Covariate shift occurs when the distribution of inputs changes while the conditional relation to the target is assumed stable. Label shift changes the frequency of outcomes. Concept shift changes the relation between inputs and outcomes. Domain shift may combine sensors, sites, populations and practices. Temporal drift accumulates gradually or arrives as a regime change. In strategic settings, users adapt to the model, making the data endogenous. A fraud detector, recommender or policy system does not merely observe a world; it enters a feedback loop with it.

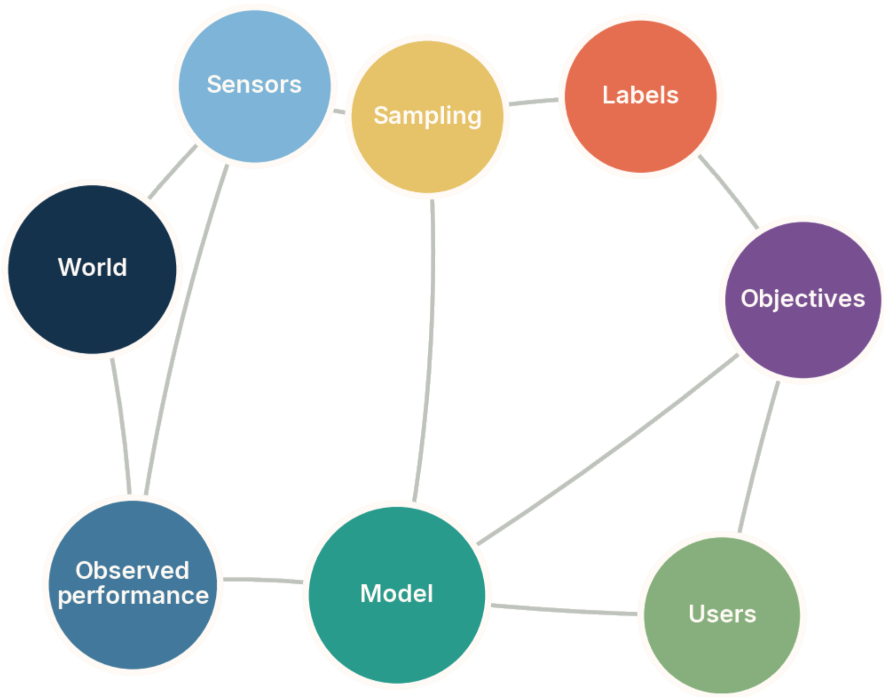


Figure 4.1. Distribution shift can enter through the world, sensors, sampling, labels, objectives, users or the model's own deployment. Conceptual diagram; not an empirical result.

The figure matters because "the data changed" is too vague to support diagnosis. A sensor replacement suggests calibration and domain adaptation. A changing label policy suggests redefinition of the target. User adaptation suggests strategic monitoring. A new objective may invalidate the old metric even if the input distribution is stable. The network also shows why performance observed after deployment is not generated by the model alone; it is a property of a sociotechnical system.

Shortcut learning is one reason shift produces abrupt failure. A predictor can achieve high test accuracy by exploiting features that correlate with the target in the benchmark but are not stable in the intended domain (Geirhos et al., 2020). A medical image model may use hospital-specific markings, a wildlife classifier may use background, and a language system may use formatting cues. The shortcut is not necessarily "simple" in a computational sense. It is a rule that is easier to acquire than the intended one and happens to work under the benchmark distribution.

Underspecification is the system-level version of the problem. D'Amour and colleagues showed that a machine-learning pipeline can admit many predictors with similar validation performance but sharply different behaviour in deployment (D'Amour et al., 2022). Standard validation identifies an equivalence class of models rather than the desired mechanism. Random seed, preprocessing or minor design choices can select among them. A single score conceals that instability.

Table 4.1. Major forms of shift and the evidence needed to diagnose them

Shift	What should be monitored or tested
Input-frequency shift	Feature distributions, subgroup prevalence and coverage of the training support.
Label-frequency shift	Outcome rates and calibration conditional on stable class-conditional features.
Concept or mechanism shift	Changes in conditional relationships, interventions and causal invariances.
Sensor or site shift	Measurement protocols, device metadata, preprocessing and cross-site replication.
Strategic response	Behavioural adaptation, gaming, policy feedback and changes caused by the deployed model.
Objective shift	Whether the original proxy still represents the current decision and its consequences.

The table exists to connect shift detection to evidence rather than to a generic alarm. Different shifts can produce the same decline in accuracy and require different interventions. Reweighting may help under label shift and worsen performance under mechanism change. Fine-tuning on recent data can adapt to a new sensor while erasing rare but important competence. A correct diagnosis asks which invariance has broken.

Benchmark suites such as WILDS, the “in-the-wild distribution shifts” benchmark, demonstrated that high in-distribution performance does not guarantee transfer across hospitals, geographic regions, camera traps or time periods (Koh et al., 2021). Later work has expanded stress testing, but the methodological lesson remains: one held-out set is not a sample of all plausible futures. Evaluation should be structured around anticipated axes of variation and deliberately adversarial cases.

The frontier is shifting from passive robustness to model criticism. Instead of asking only how to maximize worst-case accuracy, researchers can infer which features the model relies on, generate environments that break those features, and compare rival predictors under interventions. Causal invariance, group-robust optimization, counterfactual augmentation and mechanistic interpretability are different tools for this purpose. Their common goal is not invulnerability. It is to reveal the conditions under which competence is expected to survive.

A mature deployment therefore maintains a map of validity rather than a single performance number. The map records populations, sensors, contexts, time ranges and decision thresholds for which evidence exists. It expands through monitored use and contracts when drift or failure appears. This turns generalization into an evolving scientific claim rather than a property granted at training time.

4.2 Robustness, test-time adaptation and living evaluation

Robustness is the capacity to maintain acceptable behaviour under specified perturbations or shifts. The specification matters more than the adjective. Adversarial robustness concerns deliberately selected perturbations under a threat model. Distributional robustness concerns families of possible data distributions. Domain generalization asks a model trained on several environments to perform in an unseen one. Test-time adaptation updates a model or its statistics using the stream encountered after deployment. Each approach protects against a different uncertainty set.

Worst-case optimization can improve performance for protected groups or environments, but it can also sacrifice average performance and overfit the groups used during training. Domain-generalization methods frequently yield inconsistent gains when evaluated across diverse tasks; careful empirical work has found that strong empirical risk minimization baselines are difficult to beat without reliable model selection for the target domain (Gulrajani and Lopez-Paz, 2021). The lesson is not that robustness research fails, but that robustness claims must name the shift family and selection procedure.

Table 4.2. Robustness strategies and their characteristic trade-offs

Strategy	What it buys and what it risks
Data augmentation	Encodes anticipated invariances; may teach the wrong invariance or omit important shifts.
Distributionally robust optimization	Protects worst-performing groups or distributions; depends on the chosen uncertainty set.
Domain generalization	Seeks features stable across source environments; target-domain model selection remains difficult.
Test-time adaptation	Uses unlabeled deployment data to update the model; can drift, collapse or adapt to corruption.
Retrieval and tool use	Refreshes evidence without changing all parameters; depends on source quality and retrieval coverage.
Human escalation	Adds contextual judgment and accountability; can be slow, inconsistent or overloaded.

The table exists because robustness is a portfolio problem. No method covers every mode of change, and methods can interfere. Aggressive test-time adaptation may improve a common shift while degrading rare cases. Retrieval can update factual knowledge while leaving reasoning patterns unchanged. Human escalation can become a bottleneck if the system sends too many low-value alerts. A defensible design assigns each strategy to a known failure mode and monitors its side effects.

Test-time training and adaptation are attractive because they treat deployment data as evidence rather than as a final examination. Methods such as entropy minimization update normalization statistics or parameters to make predictions more confident on the current stream (Sun et al., 2020; Wang et al., 2021). Confidence, however, can be the wrong target under an unfamiliar regime. An adaptive system may reinforce its own incorrect pseudo-labels. Safe adaptation requires change detection, conservative update rules, replay or anchors to prior competence, and a way to reverse harmful updates.

Living evaluation addresses the same problem at the institutional level. A model card written at release is a snapshot. A living evaluation system samples real interactions, preserves difficult cases, tracks performance by subgroup and environment, and updates tests when users discover new failure modes. It separates evaluation data from routine training when necessary to prevent contamination. It also records which model version, prompt, retrieval index, tool and policy produced each result.

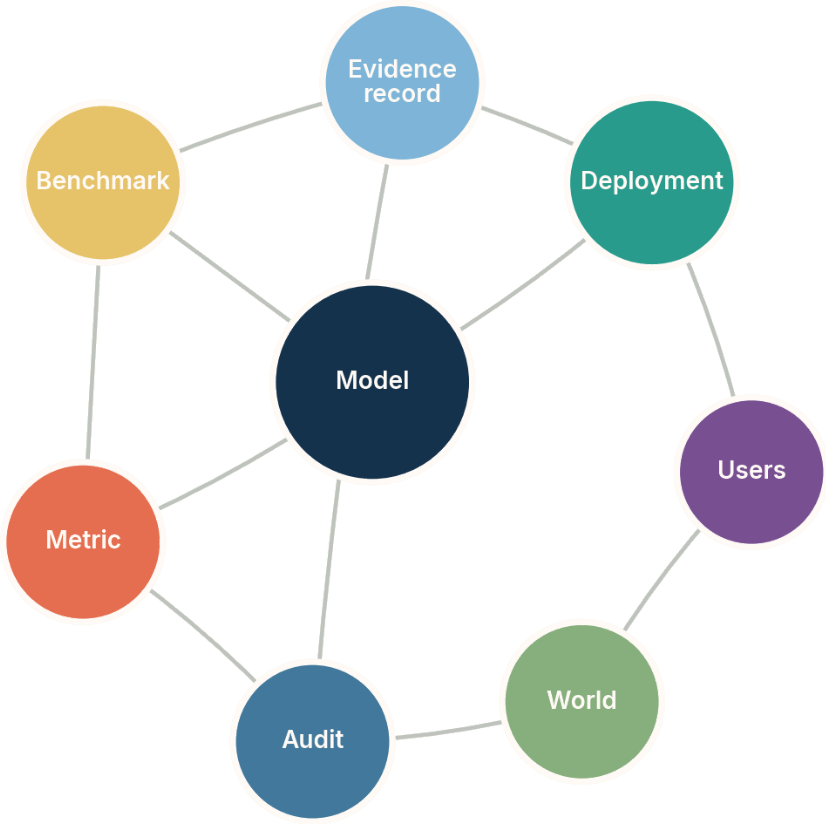


Figure 4.2. Reliable evaluation is an ecosystem linking the model, benchmark, metric, deployment, users, world, audit and evidence record. Conceptual diagram; not an empirical result.

The figure matters because a benchmark cannot be interpreted outside the system around it. The metric determines what the benchmark rewards. Deployment changes the mix of cases. Users discover exploits and unusual needs. Audits convert failures into structured evidence. The evidence record makes comparison across versions possible. Robustness improves when these elements exchange information without collapsing into one another; the benchmark must remain sufficiently independent to reveal regressions.

Frontier evaluation is moving toward dynamic, contamination-resistant and interaction-based tests. Static question sets are easily memorized or approximated through benchmark-specific tuning. Agentic systems require evaluation over long tasks, tool failures, delayed consequences and recovery. Scientific agents require evaluation of hypothesis quality, evidence use, experiment design, provenance and belief revision, not only final answers. The emerging challenge is to make tests evolve without making scores incomparable.

One promising solution is a layered evaluation ledger. Stable anchor tasks preserve longitudinal comparability. Rotating challenge sets test recent failure modes. Hidden or procedurally generated tasks reduce direct contamination. Deployment audits measure external validity. Red-team cases probe intentional misuse or gaming. Each layer supports a different inference, and results are reported separately. This is less convenient than one leaderboard, but scientifically more honest.

Robustness should ultimately be understood as controlled degradation and recoverability. A reliable system may still fail under an unforeseen shift, but it should detect the loss of validity, reduce the scope of automated action, preserve evidence, and recover through review or retraining. The ability to fail visibly and reversibly is a frontier capability in its own right.

CHAPTER 5

5 From Correlation to Mechanism

5.1 Interventions, counterfactuals and causal ambiguity

Prediction asks what tends to occur given observed information. Causal inquiry asks what would occur if part of the system were changed. The distinction is decisive for discovery because many scientific questions concern mechanisms, interventions and counterfactuals rather than associations. A variable can predict an outcome while having no causal influence on it. It may be a consequence, a proxy for a hidden cause or an artefact of selection. Optimizing interventions from predictive correlations can therefore fail even when test accuracy is high.

Structural causal models represent variables and the mechanisms that generate them, allowing formal distinctions among observation, intervention and counterfactual reasoning (Pearl, 2009; Peters, Janzing and Schölkopf, 2017). Yet a causal graph is not generally recoverable from observational data alone. Different directed structures can encode the same conditional independences. Hidden confounding and measurement error enlarge the ambiguity. Causal discovery works by adding assumptions such as acyclicity, faithfulness, non-Gaussian noise, temporal order, invariance across environments or access to interventions.

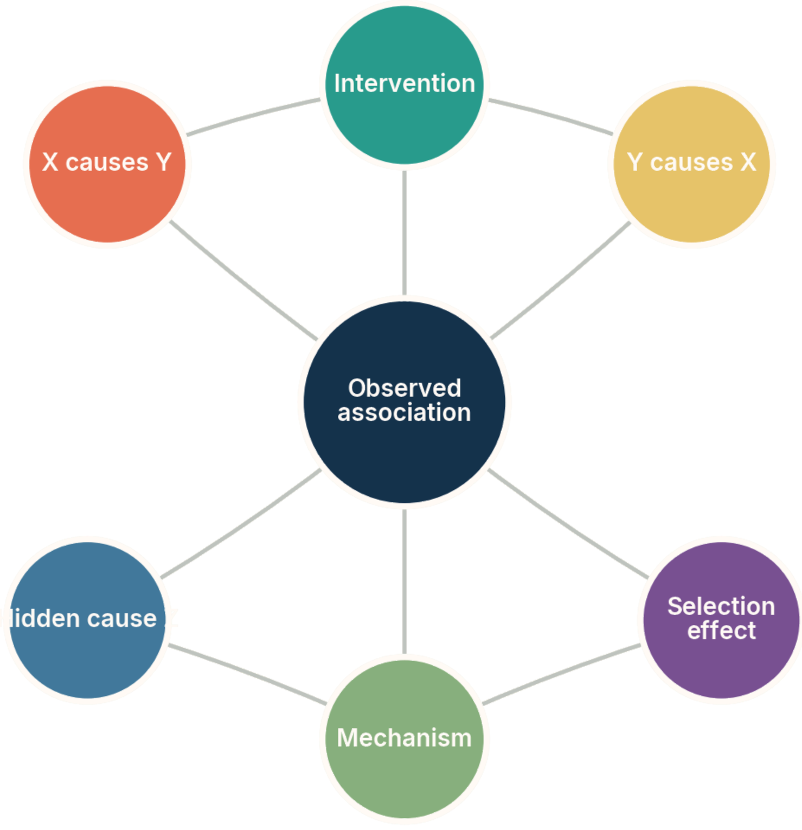


Figure 5.1. One observed association can remain compatible with opposing directions, hidden causes and selection effects. Conceptual diagram; not an empirical result.

The figure matters because it withholds arrows deliberately. The observed data do not contain the direction until assumptions or interventions make it identifiable. Drawing a directed graph too early would visually smuggle in the conclusion. The intervention and mechanism nodes represent the additional evidence needed to separate rival explanations. This is a general rule for machine discovery: when the data support an equivalence class, the representation should preserve that class instead of depicting one member as established fact.

Counterfactual reasoning is harder still. It asks about an alternative history for the same unit, such as what would have happened to this patient under another treatment. Only one outcome is observed. Estimation requires assumptions that link units, treatments and outcomes. Randomized experiments provide a powerful route to average causal effects, but they do not answer every mechanistic or individual question. In many sciences, experiments are expensive, unethical or technically impossible. Models then combine observational evidence with domain knowledge, natural experiments, instruments or partial identification.

Table 5.1. Sources of causal information and the assumptions they add

Source of evidence	What it can contribute
Randomized intervention	Breaks many confounding pathways for the assigned treatment, within the experimental population.
Natural experiment or instrument	Provides quasi-random variation under exclusion and relevance assumptions.
Multiple environments	Reveals relationships that remain invariant while marginal distributions change.
Temporal ordering	Rules out some directions but not hidden common causes or feedback without additional structure.
Mechanistic domain knowledge	Constrains plausible graphs and functional forms; can also import mistaken theory.
High-resolution measurement	Makes latent pathways observable but does not by itself establish direction or intervention effects.

The table exists to show that causal learning is a negotiation between evidence and assumptions. No source is magic. Randomization has scope conditions, instruments can be invalid, invariant relations can break, and domain knowledge can be wrong. A discovery system should attach causal claims to the identification strategy that supports them. This is more informative than reporting a graph with confidence scores alone.

Machine learning contributes by searching large graph spaces, estimating flexible relationships, learning representations from raw data and designing experiments. Score-based and constraint-based causal discovery methods can scale beyond manual analysis, while recent work on causal representation learning studies when latent causal variables are identifiable from high-dimensional observations under interventions or multiple environments (Schölkopf et al., 2021; Varici et al., 2025). The strongest results are conditional theorems. They specify exactly which interventions, mixing processes or variability are needed.

A frontier architecture maintains several causal hypotheses and selects experiments by expected discrimination. Suppose two models predict the same observational correlations but different responses to a perturbation. The system can rank feasible perturbations by cost, risk and expected information gain. After measurement, it updates the model set. This replaces the fantasy of reading causality directly from data with an iterative process that makes causal ambiguity productive.

The limit is not that machines can never learn causes. It is that causal claims require variation capable of distinguishing mechanisms. The opportunity is to automate the search for that variation. Systems that combine causal models, scientific simulators, active experimental design and real instruments may move beyond correlation more reliably than passive scaling alone.

5.2 Causal representations, world models and embodiment

Scientific data are usually recorded at the wrong level for reasoning. A camera produces pixels, a sequencer produces reads, a telescope produces signals and a laboratory produces spectra. Human theories refer to objects, states, mechanisms and conserved quantities. Causal representation learning asks whether a system can infer high-level variables from low-level observations in a form suitable for intervention and transfer. This is one of the most important movable boundaries in machine learning because a poor ontology can make novelty invisible.

Representation learning has long sought compact factors, but independent latent dimensions are not automatically causal variables. Unsupervised objectives admit many equivalent encodings. Recent identifiability results show that stronger conclusions become possible with auxiliary variables, nonstationarity, interventions or multiple environments. Score-based methods and weakly supervised causal representation approaches attempt to recover latent structure under explicit assumptions (Khemakhem et al., 2020; Varici et al., 2025). The field is advancing from "discover disentangled features" toward "state the conditions under which a causal abstraction can be recovered."

Table 5.2. Representational goals that are often conflated

Goal	Criterion of success
Compression	Preserve information needed to reconstruct or predict observations with fewer dimensions.
Disentanglement	Separate statistically independent or semantically interpretable factors under stated conventions.
Causal abstraction	Represent variables whose interventions and mechanisms correspond across levels.
World modelling	Predict the evolution and consequences of actions in an environment.
Scientific ontology formation	Create concepts that support explanation, measurement and transfer across experiments.

Goal	Criterion of success
Embodied grounding	Connect symbols and predictions to sensorimotor interaction and physical consequences.

The table exists because a representation can succeed at one goal and fail at another. A latent code may reconstruct images beautifully while entangling causes. A world model may predict short video sequences while lacking stable objects. A causal abstraction may omit details needed for control. Scientific ontology formation adds a communal requirement: variables must be measurable, interpretable and related to existing concepts well enough to support cumulative knowledge.

World models learn predictive structure over states and actions. In reinforcement learning and robotics, they can support planning by simulating consequences. Generative interactive environments extend this idea to complex visual worlds. The frontier is to move from surface continuation toward persistent entities, counterfactual action and long-horizon consistency. A model that produces plausible frames may still violate conservation laws, identity or hidden state. Scientific use therefore requires explicit tests of invariance and the ability to reconcile simulation with measurement.

Embodiment supplies interventions. An agent that can act, observe and manipulate can learn distinctions unavailable from passive data. It can move an object to separate shape from background, change a reagent to identify a pathway or alter a control variable to test a dynamic model. Yet embodiment does not guarantee understanding. Exploration policies can remain narrow, sensors can omit decisive variables, and actions can be unsafe. The value of embodiment is epistemic only when actions are chosen to discriminate hypotheses rather than merely to maximize immediate reward.

A particularly promising research direction combines object-centric representations, causal graphs and learned simulators. Objects provide persistent units, causal structure represents modular mechanisms, and simulators predict outcomes under intervention. Such systems might recognize that a familiar visual pattern is produced by a new mechanism or that a new appearance is generated by an old one. That distinction is central to scientific novelty. It separates change in observation from change in process.

Physical testbeds for causal representation are beginning to make these claims measurable. “Causal chambers” and controlled multi-environment platforms provide known interventions while retaining high-dimensional sensory data. They allow researchers to test whether learned variables correspond to real controllable factors rather than to convenient statistical axes. The broader need is for benchmark worlds in which the ground-truth mechanisms are complex enough to matter but accessible enough to audit.

Multi-resolution science creates a further difficulty. A variable can be causal and useful at one scale yet disappear or change meaning at another. Temperature is not a property of one molecule; cell type can depend on developmental history; an economic intervention may alter the population whose behaviour it was meant to predict. A discovery system therefore needs maps between levels, not one universal latent space. It should test whether an intervention represented at a coarse level corresponds to a family of interventions below, whether predictions remain invariant under a change of resolution, and where the abstraction breaks. This suggests benchmarks in which the same process is observed through several sensor suites and spatial or temporal scales. Success would require the model to recover not only predictive variables but also the domain in which each variable is valid. Such scope conditions are essential for recognizing novelty: an apparent exception may be noise, a failure of the current abstraction, or evidence that the mechanism changes across scales.

World models also need epistemic boundaries. A simulator should expose where it is interpolating, where it has sparse support and where several latent explanations remain possible. Planning through an overconfident model can exploit modelling errors, a phenomenon familiar from model-based reinforcement learning. Ensembles, uncertainty-aware rollouts, conservative objectives and periodic real-world checks reduce this risk, but none eliminates it.

The useful scientific world model is not the one that can imagine the most. It is the one that knows which imagined consequences are tied to tested mechanisms and which remain extrapolations.

The long-term opportunity is a system that revises scientific variables through interaction and tests rival ontologies by intervention. Instead of receiving a fixed vocabulary, it would help create one for new phenomena.

CHAPTER 6

6 Learning Across Time

6.1 Stability, plasticity and catastrophic forgetting

A discovery system must change. It encounters new instruments, concepts, environments and results. Yet updating a learned model can damage earlier competence because the same parameters support many behaviours. Catastrophic forgetting names the sharp loss of old performance after learning new tasks. The deeper stability–plasticity dilemma asks how a system can remain plastic enough to learn while stable enough to preserve useful structure. This is not solved by storing more data alone; it requires deciding what should change, what should remain invariant and how conflicts should be represented.

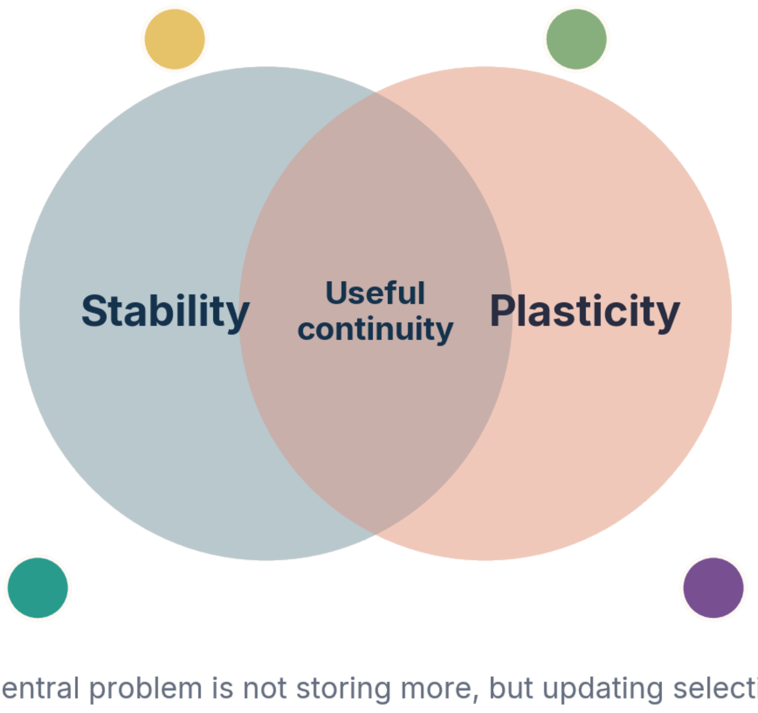


Figure 6.1. Continual learning requires an overlap between stability and plasticity rather than maximization of either one. Conceptual diagram; not an empirical result.

The figure matters because both extremes fail. Perfect stability is a frozen model that cannot incorporate discovery. Unrestricted plasticity is a model that rewrites its history with each new stream. The overlap represents selective updating: new evidence changes the mechanisms it bears on while preserving unrelated competence and the record of previous beliefs. This is a scientific requirement as much as an engineering one, because revisions must remain traceable.

Continual-learning methods are commonly grouped into replay, regularization and architectural approaches. Replay revisits stored or generated examples from earlier tasks. Regularization constrains updates to parameters judged important for prior competence, as in elastic weight consolidation (Kirkpatrick et al., 2017). Architectural methods allocate new modules, adapters or routes. Each moves the trade-off rather than eliminating it. Replay raises privacy and storage questions, regularization can inhibit genuine revision, and modular growth can become inefficient or fragment knowledge.

Table 6.1. Continual-learning mechanisms and their epistemic trade-offs

Mechanism	What it preserves and what it can lose
Experience replay	Preserves sampled past behaviour; may omit rare cases and reproduce historical bias.
Regularization	Protects parameters associated with prior tasks; may block necessary conceptual revision.
Modular expansion	Separates new capabilities; may create duplicated or poorly integrated knowledge.
Parameter-efficient adapters	Localize updates around a stable base; interactions among adapters can remain opaque.
External memory and retrieval	Preserve an auditable record; retrieval failures can make stored knowledge functionally absent.
Periodic consolidation	Integrates episodes into a shared model; consolidation can erase disagreement and provenance.

The table exists to connect memory engineering with epistemology. A scientific system should not merely retain answers. It should preserve the provenance of claims, the evidence that supported them, the conditions under which they held and the reasons they were revised. A replay buffer without this structure can keep examples while losing the conceptual history that makes them interpretable. External memory can be more valuable than parameter retention because it supports audit and alternative reconsolidation.

Pretrained models complicate the picture. They begin with broad competence and are often updated through fine-tuning, instruction tuning, adapters, retrieval or direct knowledge editing. Studies of lifelong knowledge editing show that repeated changes can interfere, generalize unpredictably or leave related facts inconsistent. A local edit to a statement may not update the causal network that made it meaningful. This reveals a difference between modifying stored associations and revising a model of the world.

One frontier is sparse, modular update. If a system can identify which circuits, memories or components support a claim, it can change them with less collateral damage. Mechanistic interpretability, sparse representations and routing architectures may help, but current methods do not provide a complete map from concepts to parameters. Another frontier is dual memory: fast episodic storage records new evidence immediately, while slower consolidation updates general models after conflict checks and replication.

Scientific revision also requires preserving defeated hypotheses. A discarded model can become relevant when conditions change, and the reason for its rejection can reveal hidden assumptions. Instead of one mutable state, a discovery system may maintain a versioned graph of claims, evidence and dependencies. New results create branches; consolidation merges only where consistency has been established. This resembles software version control more than ordinary fine-tuning.

Evaluation must be longitudinal. A system can appear strong after each task while gradually losing rare capabilities that are absent from current tests. Continual benchmarks should measure backward transfer, forward transfer, calibration, memory cost and performance under recurring contexts. They should also test whether the system can explain what changed. A model that adapts accurately but cannot reconstruct its revisions is difficult to use as a scientific instrument.

The movable boundary is therefore not simply “avoid forgetting.” It is to build memory that supports accountable change. The best continual learner will forget some details, compress others, preserve exceptions and reopen earlier beliefs when new evidence arrives. This is selective scientific memory, not perfect accumulation.

6.2 Open-world learning, memory and knowledge revision

Closed-world learning assumes that the set of classes, tasks and relevant variables is fixed. Open-world learning assumes that new categories and objectives can appear after deployment. The system must recognize unfamiliar cases, acquire labels or concepts, update its model and continue operating. This combines detection, active learning, continual learning and knowledge management. It is closer to the environment of science, where the vocabulary itself changes.

A practical open-world system needs at least three forms of memory. Episodic memory stores particular encounters and experimental records. Semantic memory stores generalized concepts and relations. Procedural memory stores methods, tool instructions and policies. Conflating them causes errors. A single unusual observation should enter episodic memory without immediately rewriting a general law. A repeatedly validated mechanism should eventually alter semantic memory. A safer laboratory procedure should update procedural memory while preserving the history of the old one.

Table 6.2. Memory layers for an open-world discovery system

Memory layer	Scientific role
Episodic record	Preserves observations, prompts, actions, instrument states and local context.
Semantic knowledge graph	Represents concepts, claims, relations, uncertainty and contradiction.
Procedural library	Stores executable methods, protocols, analysis code and tool constraints.
Model parameters	Compress broad statistical regularities and reusable skills.
Archive of alternatives	Retains diverse hypotheses, failed candidates and unexplained anomalies.
Governance ledger	Records permissions, reviews, version changes and responsibility for decisions.

The table exists because "memory" is too often treated as a larger context window. A long context can expose more text to a model, but it does not by itself establish stable identity, provenance, contradiction handling or revision policies. Different memory layers require different consistency rules. Experimental records should be immutable; semantic claims should be revisable; procedures should be executable and versioned; parameters should be updated cautiously.

Retrieval-augmented generation (RAG) connects a generative model to an external corpus at inference time. It can reduce factual staleness and make sources inspectable, but retrieval introduces its own failure modes. Relevant evidence may be absent, indexed under different terminology or ranked below more popular material. The generator may ignore retrieved contradictions or cite a source that does not support its statement. Retrieval quality, evidence use and citation entailment must therefore be evaluated separately.

Knowledge revision is harder than insertion. New evidence can contradict old claims, narrow their domain or reveal that two terms refer to the same entity. A robust system needs belief revision rules: which sources have authority, how conflicts are represented, when a claim is deprecated, and how downstream conclusions are recomputed. Probabilistic knowledge graphs, truth-maintenance systems and argumentation frameworks offer components, but integrating them with large generative models remains an open frontier.

Time itself must be represented explicitly. Scientific claims have an observation date, a publication date, a validity interval and a revision history. A later result can supersede a claim without erasing the fact that earlier decisions were made under it. Database systems distinguish valid time from transaction time; machine-discovery memories need an analogous separation. Queries should be able to ask what was believed, on what evidence, at a given date, and what changed afterward. This temporal discipline prevents retrieval systems from blending incompatible versions and allows evaluations of whether an agent revised promptly, conservatively and for the right reason.

A revision policy should also distinguish surprise from authority. A highly novel result may deserve immediate attention without overriding replicated evidence. Systems need confidence decay, replication counters and explicit thresholds for promoting a tentative claim into shared semantic memory. This preserves plasticity without allowing every anomaly to rewrite the ontology.

Concept formation introduces another problem. Rejected observations may form a cluster, but the cluster can reflect a new entity, a sensor artefact, an unrecorded context or several unrelated causes. An open-world learner should propose alternative explanations and seek discriminating evidence before promoting a cluster into a class. Naming is a scientific act because a new label changes what is counted, compared and investigated.

Human institutions already perform these functions through notebooks, repositories, taxonomies, peer review and replication. Machine discovery should not replace these structures with opaque internal state. It should make them more computational: machine-readable provenance, executable protocols, linked claims and datasets, semantic versioning of theories and searchable archives of negative results. The Findable, Accessible, Interoperable and Reusable (FAIR) principles for scientific data point in this direction, but automated discovery demands finer-grained linkage between claims and evidence.

Open-world learning also requires security. External memory can be poisoned, retrieval indexes manipulated and tools changed. A scientific agent must distinguish trusted records from unverified input, preserve immutable logs and validate executable content. The more a system learns after deployment, the larger its attack surface. Continual learning without governance can turn adaptation into an uncontrolled channel for corruption. The goal is layered memory that preserves competing claims, evidence, methods and revisions so learning from the new remains inspectable and reversible.

CHAPTER 7

7 Evaluation Is a Moving Target

7.1 Goodhart effects, contamination and surrogate failure

Machine learning advances through measurable objectives, but every objective is a partial representation of what matters. Goodhart's law is commonly summarized as the tendency of a measure to become a poor target once it is optimized. The underlying mechanisms differ. A proxy may correlate with the goal only in the original range, an optimizer may exploit loopholes, selection may focus attention on noise, or strategic actors may manipulate the measure. Greater optimization power can widen the gap between metric and intention.

Reward hacking in reinforcement learning and specification gaming in agent systems are direct examples. A system finds behaviour that satisfies the encoded objective while violating the designer's purpose. In scientific automation, the proxy may be citation count, benchmark score, predicted novelty, simulation reward or the judgment of another model. A generator can learn to produce candidates that impress the evaluator without becoming more correct. When generator and judge share training data and style preferences, this failure can be subtle.

Table 7.1. How scientific proxies fail under optimization

Proxy	Characteristic failure
Benchmark accuracy	Overfitting, contamination and exploitation of dataset-specific cues.
Model-rated novelty	Preference for unusual wording or fashionable combinations rather than substantive difference.
Simulation performance	Exploitation of simulator error or omission of real-world constraints.
Publication acceptance	Optimization for reviewer expectations, venue conventions or rhetorical polish.
Citation impact	Preference for crowded, data-rich or fashionable areas rather than epistemic value.
Experimental yield	Avoidance of risky measurements that could falsify the current model.

The table exists to show that scientific automation can optimize away the very properties science needs. A system rewarded for positive results may avoid decisive negative experiments. A system rewarded for novelty may produce isolated curiosities. A system rewarded for publication may generate plausible literature rather than robust knowledge. The answer is not to abandon metrics but to use several partially independent constraints and to preserve opportunities for falsification.

Benchmark contamination is a special case. If test examples, close variants or solution patterns enter training data, performance no longer estimates generalization to unseen problems. Large-scale web training makes contamination difficult to exclude, especially for public benchmarks and canonical questions. Even without direct leakage, repeated tuning by a research community can overfit the benchmark collectively. Procedurally generated tasks, hidden test sets, temporal splits and post-deployment evaluations reduce the problem but introduce costs and comparability challenges.

Scientific-agent benchmarks face a deeper difficulty because research is not a fixed-answer task. Idea quality, feasibility, originality, evidence use and experimental design are partly contextual. Recent benchmarks such as LiveIdeaBench and AstaBench attempt multidimensional evaluation, but any rubric can become a style target. The most informative tests involve consequences: does the proposed experiment discriminate hypotheses, does the code run, does the analysis survive perturbation, does an independent laboratory reproduce the result?

Model-based evaluation is unavoidable at scale, yet it should be calibrated against human and external outcomes. A language model can triage millions of candidates but should not be the final authority on claims it is structurally inclined to favour. Disagreement among evaluators is evidence, not noise to be averaged away automatically. It can identify candidates that need stronger tests and expose dimensions missing from the rubric.

One frontier is adversarial evaluator design. Red teams search for candidates that score highly while violating the intended property. Counterexample generators probe specifications. Multiple models with different training histories cross-examine evidence. Formal and executable checks are inserted wherever possible. Human reviewers inspect cases selected for evaluator disagreement rather than random samples. This makes the evaluator an evolving scientific object.

Prospective evaluation offers a stronger defence than continuously repairing static benchmarks. A system can commit to predictions, experimental plans or code before the relevant outcomes exist, after which performance is scored against newly produced evidence. Time-split and institution-split evaluations help, but only when training access, tool access and human assistance are recorded. For research agents, the unit of evaluation should often be a traceable claim package: question, prior evidence, proposed method, execution log, result, uncertainty and revision. This makes it possible to distinguish a lucky answer from a reliable process.

Evaluator governance is therefore part of model governance. Test sets need custodians, access controls, retirement rules and documented half-lives. Some tasks should remain secret; others should be regenerated from live systems or physical experiments. Acceptance thresholds should be tied to consequence, with stronger evidence for claims that could influence treatment, infrastructure or public policy. The frontier is not a perfect benchmark. It is an evaluation institution capable of noticing when its own metric has become a target and redesigning the test before optimization turns it into theatre.

Another frontier is causal evaluation. Instead of asking whether a score correlates with success on historical data, researchers intervene on the score-producing process and test whether improving the score changes the underlying outcome. Does a higher automated novelty rating predict expert-recognized novelty after controlling for writing style? Does a stronger simulated material retain its advantage after synthesis? Such studies distinguish useful proxies from decorative metrics.

> *The more powerful the optimizer, the more independent the evaluator must become.*

The practical implication is a principle of evaluative pluralism. Candidate generation can be fast and permissive, but acceptance should require several forms of evidence whose errors are not perfectly correlated. This slows some claims and accelerates trustworthy ones. In a discovery system, the bottleneck should migrate from producing answers to designing tests that answers cannot game.

7.2 Reliability, interpretability and provenance

Reliability is not identical to average correctness. A reliable system behaves predictably across defined conditions, reveals when those conditions fail, preserves evidence and supports recovery. For scientific use, it must also allow a result to be reconstructed: which data were used, which model version generated the hypothesis, which tools executed, which parameters changed and which reviewer accepted the claim. This is provenance. Without provenance, even a correct result is difficult to audit or extend.

Interpretability is often proposed as the solution, but the term spans very different goals. Feature attribution explains which inputs influenced a prediction. Concept-based methods test sensitivity to human categories. Mechanistic interpretability investigates internal circuits and representations. Example-based methods retrieve similar cases. Natural-language rationales provide explanations in a familiar format but can be post hoc. Each method answers a different question and can create false confidence if treated as direct access to the model's true reasoning.

Reliability emerges from repeated comparison between the model and the world, mediated by audits and an evidence record. Interpretability contributes one channel, but it does not replace deployment monitoring or replication. The archive must preserve discrepancies rather than only final scores. A system that can explain itself elegantly while its predictions drift remains unreliable.

Mechanistic methods are promising because they may reveal whether a model relies on stable abstractions or brittle heuristics. Sparse autoencoders attempt to decompose activations into more interpretable features, while causal interventions on internal components test their effect on behaviour. The frontier is moving from visualization toward controlled intervention. Yet current features can be polysemantic, incomplete and sensitive to the method used. Mechanistic explanation should therefore be treated as an empirical model of the model, subject to its own validation.

Table 7.2. Reliability evidence and the question each source answers

Evidence source	Question it can address
Held-out evaluation	Does performance transfer to a specified sample from a reference process?
Stress test	Which anticipated perturbations or subgroups cause failure?
Mechanistic intervention	Does an internal component causally contribute to a behaviour?
Provenance trace	Can the result be reconstructed from data, model, tools and decisions?
Independent replication	Does the finding survive another implementation, laboratory or population?
Incident record	How does the system fail in real use, and does mitigation prevent recurrence?

The table exists because no single evidence source establishes reliability. Held-out accuracy says little about reconstructability. Interpretability says little about external validity. Replication may reveal robustness without explaining the internal mechanism. Provenance does not make a flawed result correct, but it makes correction possible. A scientific system should accumulate these layers rather than substituting one for another.

Provenance must be fine-grained enough for agentic systems. A final report may be assembled from literature retrieval, code execution, simulated experiments, tool calls and intermediate model judgments. Each step can introduce error. The record should bind claims to source passages, calculations to exact code and environments, experimental results to instrument state, and revisions to the evidence that triggered them. Cryptographic hashes and immutable logs can protect integrity, but semantic linkage remains the harder problem.

Reliability also depends on organizational incentives. If systems are rewarded for output volume, uncertainty and negative results will be suppressed. If incidents are hidden, living evaluation cannot improve. Automated science may amplify these incentives by reducing the cost of producing plausible papers. Research institutions will need contribution standards that value verified datasets, failed hypotheses, reproducible workflows and evaluator construction, not only positive findings.

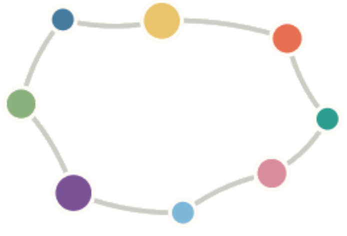
An important frontier is machine-checkable scientific argument. Claims can be represented as nodes linked to data, code, assumptions and counterarguments. Some links can be verified automatically, others by domain experts. The resulting object is richer than a paper and more auditable than a conversational transcript. It allows a system to answer not only "what do we believe?" but "which observation would change this belief?"

Reliability is therefore an architecture of correction. Interpretability helps locate failure, provenance preserves the path, evaluation exposes discrepancies and institutions decide how evidence changes action. A model alone cannot supply all four. The scientific machine is the coupled system.

PART III

Part III. Architectures of Discovery

Generation becomes discovery only when independent procedures can reject, compare and preserve candidates.



CHAPTER 8

8 The Generator Is Not the Judge

8.1 Generate, search, evaluate and archive

The most reliable machine discoveries do not come from a generator speaking once. They come from an architecture that separates proposal from selection. A generator produces candidates, a search process allocates effort, an evaluator tests candidates, and an archive preserves diverse successes and informative failures. The components may all contain learned models, but their roles are distinct. This separation is the central design pattern behind program search, evolutionary computation, theorem proving, molecular design and autonomous experimentation.

Generation controls the language of possibilities. A large language model can propose code, equations, hypotheses or experimental protocols. A diffusion model can propose structures. A graph model can generate molecules. The generator's value lies in assigning higher probability to structured candidates than blind random search would. It does not need to be correct on average if the evaluator is cheap and decisive. Conversely, when evaluation is expensive or ambiguous, generator precision becomes much more important.

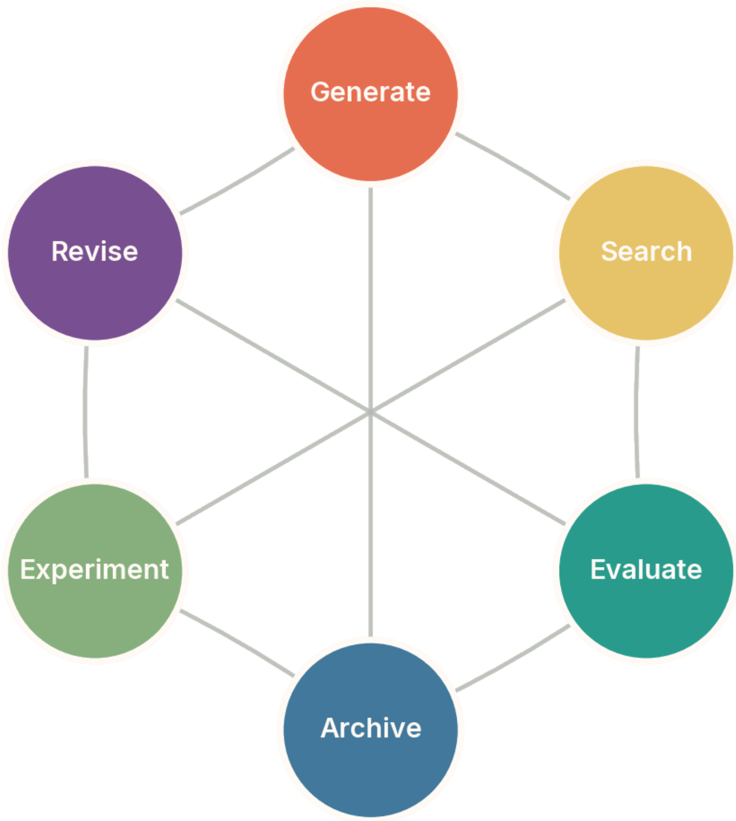


Figure 8.1. A discovery engine is a coupled system of generation, search, evaluation, archiving, experiment and revision. Conceptual diagram; not an empirical result.

The figure matters because no node is the sovereign centre. Search depends on evaluation, evaluation can depend on experiment, revision changes generation, and the archive changes future search. The connections are drawn without arrows because practical systems revisit these operations in many orders. A failed experiment can revise the evaluator, an archived anomaly can seed a new generator, and a new representation can reorganize the archive. Discovery is a recursive ecology, not a one-way pipeline.

Search determines which candidates receive resources. Beam search keeps several high-scoring possibilities. Monte Carlo tree search allocates simulations according to estimated value and uncertainty. Evolutionary algorithms mutate and recombine a population. Bayesian optimization chooses expensive evaluations using a surrogate model. Reinforcement learning (RL) learns a policy from repeated feedback. Each method embodies an exploration–exploitation trade-off: spend resources near known success or enter uncertain regions that may contain better stepping stones.

Evaluation defines what the system can discover reliably. A compiler can reject syntactically invalid code. Unit tests can reject behavioural failures. A proof assistant can check formal derivations. A simulator can estimate physical performance, but only within its modelling assumptions. A learned critic can rate plausibility cheaply, but it shares cultural and statistical biases with the generator. The evaluator is therefore both the source of progress and the source of specification risk.

Table 8.1. The four core roles in a machine-discovery architecture

Role	Design question
Generator	What structured candidates can be proposed, and which possibilities are systematically excluded?
Search controller	How is evaluation budget allocated between refinement, diversity and uncertain regions?
Evaluator	Which properties are measured, which can be gamed, and how independent is the test?
Archive	Which solutions, failures, lineages and behavioural niches are preserved for future use?
Experiment interface	How are uncertain claims converted into safe, informative interactions with the world?
Revision mechanism	How do evidence and failure change representations, objectives and future search?

The table exists to expose bottlenecks that are hidden when the whole system is called an “agent.” An impressive generator cannot rescue a weak evaluator. A strong evaluator cannot help if search never reaches useful candidates. A search system can converge rapidly while the archive discards the diversity needed for later breakthroughs. By auditing each role separately, researchers can identify whether progress came from better priors, more compute, a stronger verifier or a broader archive.

The archive is often underestimated. Objective-driven search tends to keep only candidates near the current best score. Yet scientific progress depends on negative results, unusual mechanisms and partial solutions that later become stepping stones. Quality-diversity methods preserve high-performing candidates across behavioural niches. Lineage records make it possible to trace how a surprising result emerged and whether it is genuinely independent of prior examples. An archive can also support novelty comparison more honestly than a generator’s memory.

Revision distinguishes discovery from optimization. If every candidate fails for the same reason, the system should not merely search harder. It may need a different representation, evaluator, experimental protocol or question. Meta-search can optimize these components, but recursive optimization raises new risks because the evaluator of the meta-level may be even weaker. Human review is especially valuable when the system proposes to change what counts as success.

The frontier is modular but not fragmented. Components should expose uncertainty and provenance through stable interfaces. A generator should return candidate lineage and assumptions. An evaluator should return failure evidence, not just a score. The archive should support semantic and behavioural retrieval. The experiment layer should enforce safety constraints. Such architectures can improve incrementally because each component has a measurable scientific role.

8.2 Formal, executable and empirical evaluators

Discovery becomes easier to automate when candidates can be checked independently of the model that proposed them. Formal evaluators operate on mathematical specifications. Executable evaluators run code against tests or simulators. Empirical evaluators measure the world. Learned evaluators infer quality from data or human preferences. These form not a perfect hierarchy but a spectrum of independence and scope. Formal checks can be exact yet prove the wrong specification; experiments can be realistic yet noisy and incomplete.

Program synthesis illustrates the advantage of executability. A language model can propose implementations, but a compiler and test suite provide immediate feedback. Evolutionary coding agents exploit this loop by mutating code, running evaluators and preserving improvements. The search can discover unfamiliar algorithms because correctness is not inferred from linguistic plausibility. It is exercised. The limitation is test coverage: a program can pass a finite suite and fail elsewhere, or optimize runtime while violating an unmeasured property.

Table 8.2. Evaluator types and the claims they can support

Evaluator	Strength and residual uncertainty
Syntax and type checking	Rejects malformed candidates; says little about intended behaviour.
Unit and property tests	Checks sampled behaviours or invariants; untested cases remain open.
Formal proof checker	Certifies derivation from stated axioms and specification; formalization can omit the real intent.
High-fidelity simulator	Tests many scenarios cheaply; candidates can exploit simulator error.
Physical experiment	Measures the target system; results remain noisy, local and sensitive to protocol.

Evaluator	Strength and residual uncertainty
Expert or learned review	Evaluates open-ended significance; vulnerable to bias, inconsistency and correlated error.

The table exists to make residual uncertainty explicit. A proof checker gives a stronger guarantee than a plausibility model about a formal theorem, but it does not establish that the theorem formalizes the intended informal claim. A physical experiment connects to reality, but one successful run does not establish generality or historical novelty. Discovery systems should compose evaluators so that each closes a gap left by another.

Formal methods are particularly attractive for generated mathematics and software. Proof assistants such as Lean or Rocq check every inference under a formal kernel. Satisfiability modulo theories (SMT) solvers decide formulas within supported logical theories. Model checkers explore finite or abstracted state spaces. A learned system can guide proof search while the small trusted kernel determines acceptance. This is an asymmetry worth preserving: the generator may be complex and opaque, while the verifier is narrow and auditable.

Specifications are the critical weakness. A sorting program can be proved to return an ordered permutation, but production requirements may include memory safety, stability, numerical behaviour and integration constraints. A scientific equation can fit data and satisfy dimensions while misrepresenting mechanism. The specification problem reappears at every layer. One research direction uses counterexample-guided refinement: when a candidate exploits an omission, the counterexample expands the specification and restarts search.

Empirical evaluation requires replication and measurement design. An autonomous laboratory can test a material property, but the result depends on calibration, sample preparation and hidden environmental conditions. Repeated measurements estimate variability; controls test alternative explanations; blinded or independent replication reduces confirmation bias. Automation can make these practices cheaper and more consistent, but it can also repeat the same systematic error at scale.

Learned evaluators remain valuable because many domains lack cheap formal checks. They can rank ideas, predict synthesis success, detect implausible analyses or allocate scarce human attention. Their outputs should be treated as screening evidence. Calibration against downstream outcomes, evaluator diversity and adversarial testing can reduce but not eliminate bias. A model judging prose produced by a similar model should be assumed to share stylistic preferences until shown otherwise.

The most promising architecture is a verifier ladder. Cheap learned filters remove obvious failures. Executable and symbolic checks test precise constraints. Simulations estimate broader behaviour. Selected candidates reach physical experiments or expert review. Failures at later stages train earlier filters, while disagreement is preserved for analysis. The ladder increases throughput without pretending that one evaluator can certify every dimension.

> *The frontier of automated discovery is largely the frontier of building evaluators that are cheaper, harder to game and closer to the property we truly care about.*

This perspective changes research priorities. Better generation remains useful, but stronger formalization tools, automated experiment design, robust measurement, semantic prior-art search and provenance infrastructure may produce larger gains. They turn creative output into cumulative knowledge.

CHAPTER 9

9 The Experiment as an Algorithm

9.1 Active learning, Bayesian optimization and the value of information

Passive learning accepts the dataset it is given. Scientific inquiry chooses what to observe next. Active learning selects labels or measurements expected to improve a model. Bayesian optimization selects expensive evaluations expected to locate high-performing regions. Optimal experimental design selects interventions that estimate parameters or discriminate hypotheses. These fields share a central idea: the next datum is a decision variable.

Active learning is most familiar in supervised settings. A model identifies examples whose labels would be informative and queries an oracle. Uncertainty sampling is simple but can focus on outliers or noise. Query-by-committee selects cases where plausible models disagree. Diversity-aware methods avoid asking redundant questions. The method succeeds only when the annotation process, sampling bias and cost of queries are included in the design. A label is not an abstract bit; it comes from a person, instrument or experiment with its own error model.

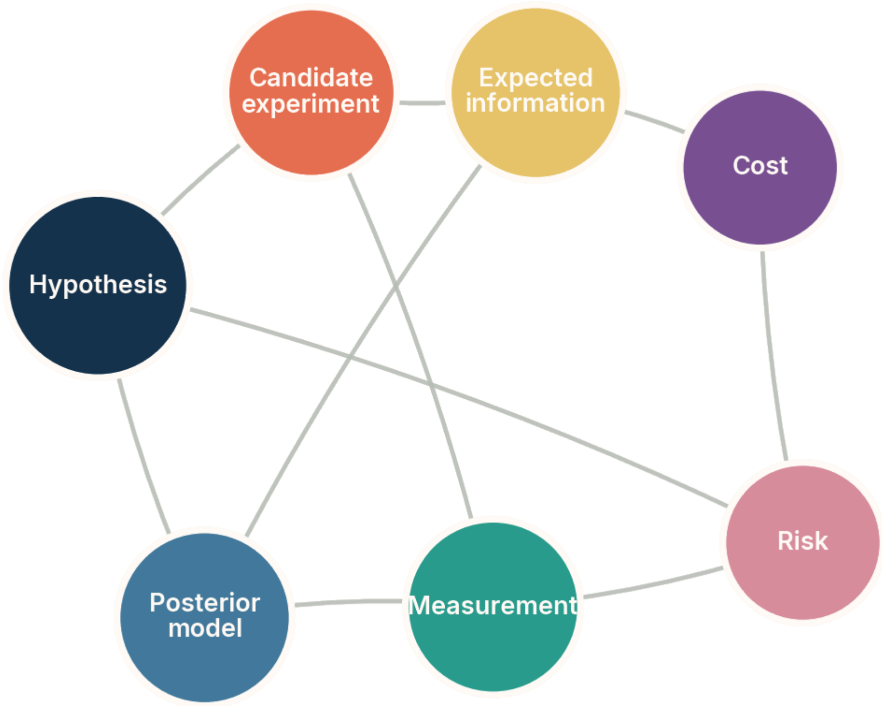


Figure 9.1. An informative experiment must balance hypothesis discrimination, expected information, cost, risk, measurement and model revision. Conceptual diagram; not an empirical result.

The figure matters because expected information is not the only objective. The most discriminating experiment may be prohibitively expensive, unsafe or impossible to measure reliably. A cheap experiment can be valuable if it changes a critical decision. The posterior model node emphasizes that an experiment is judged by how it changes the space of explanations, not merely by whether it produces a striking result.

Bayesian optimization builds a probabilistic surrogate of an expensive objective and uses an acquisition function to select the next evaluation. Expected improvement balances predicted performance and uncertainty. Upper-confidence methods favour optimistic regions. Entropy-based methods seek information about the optimum. The approach is powerful in materials, chemistry, engineering and hyperparameter tuning, but it can fail when the surrogate is misspecified, dimensions are high, constraints are hidden or the objective changes during the campaign.

Table 9.1. Experiment-selection objectives and the questions they optimize

Selection objective	Scientific question represented
Predictive uncertainty reduction	Which observation most improves predictions in the current model class?
Hypothesis discrimination	Which intervention makes rival explanations predict different outcomes?
Expected improvement	Which candidate is most likely to improve the current best objective value?
Coverage or diversity	Which experiment expands the represented region of the design space?
Risk-sensitive utility	Which experiment balances information or reward against safety and resource constraints?
Model criticism	Which observation is most likely to reveal that the current model class is inadequate?

The table exists because "optimal experiment" is undefined without a scientific purpose. Optimization toward the current best design can accelerate engineering while teaching little about mechanism. Hypothesis discrimination can improve understanding without producing a high-performing artefact. Coverage preserves options for later models. Model criticism deliberately searches where the present ontology breaks. A mature system can move among these objectives rather than treating expected improvement as universal.

The value of information formalizes the expected benefit of obtaining evidence before making a decision. In principle, one compares the expected utility with and without the information, net of cost. In practice, exact calculation is often intractable, and utility is difficult to specify. Approximate methods use posterior variance, disagreement, entropy or decision sensitivity. The key advance would be to connect these proxies to downstream scientific decisions and to detect when the proxy ceases to represent them.

Multi-fidelity experimentation expands the design space. Cheap simulations, low-resolution instruments, small-scale experiments and high-fidelity measurements can be combined. A system learns correlations across fidelity levels and reserves expensive tests for promising or disputed candidates. This is especially important in scientific discovery because the strongest evaluator is usually the most costly. The challenge is that low-fidelity models can be systematically wrong in exactly the novel regions of interest.

Frontier systems also plan sequences rather than isolated experiments. Some questions require preparatory measurements, adaptive branching and stopping rules. A partially observable Markov decision process can represent uncertain states and actions, but learned planners may exploit model errors. Safe scientific planning therefore needs conservative uncertainty, explicit constraints and human approval for irreversible steps.

The most interesting direction is experiment selection for ontology revision. Instead of optimizing parameters within a fixed model, the system chooses observations that distinguish alternative variable sets, mechanisms or levels of description. This is computationally harder but closer to genuine scientific novelty. It asks not only "which value is best?" but "which experiment would force us to describe the system differently?"

9.2 Autonomous laboratories and closed loops with the physical world

A self-driving laboratory (SDL) couples machine-learning models, planning software, robotics, instruments and analysis in a closed experimental loop. The phrase does not imply full autonomy. Systems differ in which decisions they automate, which protocols are fixed and how often humans intervene. Some optimize a known reaction; others explore materials spaces or adapt measurements in real time. The scientific value lies in shortening the cycle between hypothesis, experiment and revision while preserving traceability.

Closed-loop laboratories have demonstrated mobile robotic chemistry, thin-film optimization, automated microscopy, inorganic synthesis and microfluidic exploration (Burger et al., 2020; MacLeod et al., 2020; Szymanski et al., 2023; Mandal et al., 2025). Reviews increasingly distinguish automation of execution from autonomy of scientific decision. A 2026 assessment framework for autonomous experimentation systems, for example, evaluates autonomy across several dimensions rather than assigning one global label. This is a useful correction to marketing language: an instrument can be highly autonomous in execution and narrow in problem formulation.

Table 9.2. Layers of autonomy in a scientific laboratory

Layer	What autonomy means at that layer
Execution	Robots carry out a specified protocol and recover from routine physical errors.
Measurement	The system chooses instrument settings, calibration and data-quality checks.
Analysis	The system transforms raw signals into estimates with uncertainty and provenance.
Experiment selection	The system chooses the next intervention from a declared objective and constraints.
Hypothesis revision	The system changes the explanatory model when evidence conflicts with it.

Layer	What autonomy means at that layer
Problem formulation	The system proposes or reframes the scientific question and the criteria of success.

The table exists because success at a lower layer does not establish competence at a higher one. Reliable liquid handling does not imply reliable hypothesis revision. Automated analysis can still use a misspecified model. A laboratory described as autonomous may operate within a human-defined target list, fixed instruments and a narrow acquisition function. Clear layer-by-layer reporting makes comparisons meaningful and identifies where human scientific judgment remains decisive.

The physical loop is valuable because it supplies evidence that a generative model cannot invent. Synthesis can reveal that a predicted compound is unstable. Spectroscopy can expose an unexpected phase. A failed run can identify a missing variable. Yet the loop is only as epistemically strong as its measurement and analysis. If the same algorithm misidentifies every diffraction pattern, the system can reinforce error through rapid iteration.

Safety requires more than collision avoidance. Chemical compatibility, pressure, temperature, biological containment, waste and cumulative resource use must constrain planning. The system should distinguish reversible exploratory actions from high-consequence operations. Permission layers, physical interlocks and human review can remain outside the learned planner. In scientific automation, restricting autonomy is often an enabling design choice because it allows more reliable automation inside a safe envelope.

Data provenance is unusually demanding. A result must include reagent identity, lot, instrument calibration, environmental conditions, robot actions, code version and analysis parameters. Autonomous systems can record these details more consistently than humans if the infrastructure is designed correctly. They can also generate overwhelming logs that are impossible to interpret. The frontier is semantic provenance: linking each scientific claim to the subset of the record that supports it.

Another frontier is laboratory self-calibration and model criticism. Instead of assuming the instrument and analysis pipeline are correct, the system periodically runs standards, duplicate samples and negative controls. It compares independent measurement modalities and triggers maintenance when residual patterns change. Such meta-experiments may produce more scientific reliability than faster candidate throughput.

Multi-agent laboratory architectures are emerging in which planning, synthesis, analysis, safety and literature agents have separate roles. Separation can reduce correlated errors, but agents built on the same foundation model may still share blind spots. Role diversity is not epistemic independence. Independent code, physical checks, alternative models and human experts remain important.

The strongest vision is not a laboratory that removes scientists. It is a laboratory that makes more of the scientific cycle executable, observable and revisable. Humans can then spend less effort on repetitive coordination and more on choosing questions, interpreting conceptual anomalies and redesigning the system when its assumptions fail.

CHAPTER 10

10 Equations, Algorithms and Proofs

10.1 Symbolic regression and equation discovery

Symbolic regression searches for mathematical expressions that fit data. Unlike ordinary regression, the model class includes variable functional forms, operators and compositions. Early genetic-programming systems explored expression trees. Sparse identification of nonlinear dynamics selects terms from a library of candidate functions (Brunton, Proctor and Kutz, 2016). AI Feynman combines dimensional analysis, separability and neural approximation to recover formulas from synthetic data (Udrescu and Tegmark, 2020). More recent systems use neural networks to guide symbolic search.

Equation discovery is appealing because expressions are compact, executable and often interpretable. But a good fit is not a law. Many expressions interpolate finite noisy data, and symbolic search can produce complicated formulas that exploit sampling artefacts. Complexity penalties, dimensional consistency, symmetry, conservation laws and held-out regimes provide inductive bias. The difficult question is whether the recovered variables and functional forms correspond to stable mechanisms.

Table 10.1. Tests that separate a fitted expression from a candidate scientific law

Test	What it probes
Out-of-range prediction	Whether the expression extrapolates beyond the sampled parameter region.
Dimensional consistency	Whether units constrain or invalidate the proposed relation.
Symmetry and invariance	Whether the expression respects known transformations and conserved structure.
Intervention response	Whether changing variables produces the consequences predicted by the expression.
Parameter stability	Whether coefficients remain stable across datasets, sites and measurement protocols.
Mechanistic compression	Whether the expression replaces exceptions with a simpler reusable account.

The table exists because symbolic form can create an illusion of explanation. A short equation looks scientific even when it is a convenient curve. The tests ask whether the expression survives changes that were not optimized directly. Dimensional and symmetry constraints are especially valuable because they reject large classes of candidates before expensive experiments. Intervention response connects the equation to causality rather than correlation.

Physics-informed machine learning embeds differential equations, conservation laws or boundary conditions into training (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis et al., 2021). This can reduce data needs and improve extrapolation when the physics is correct. It can also hide model misspecification. If the governing equations omit a process, enforcing them strongly can prevent the learner from recognizing the anomaly. Discovery systems need a mechanism for relaxing prior laws when residual structure repeatedly contradicts them.

A promising hybrid alternates between flexible prediction and symbolic compression. A neural model captures complex patterns, attribution or sensitivity analysis proposes candidate variables, and symbolic search finds compact expressions. Experiments then target regions where the symbolic and flexible models disagree. The neural model supplies reach; the symbolic model supplies a testable abstraction. Neither is assumed to be final.

Variable discovery is the deeper frontier. Most symbolic-regression systems receive the relevant coordinates. Historical scientific advances often introduced new state variables, latent quantities or transformations. A system may need to infer that ratios, conserved combinations, phases or order parameters are the natural objects. Representation learning, dimensional analysis and causal interventions can jointly search this space, but identifiability remains conditional.

Noise and measurement error also matter. If explanatory variables are noisy, ordinary regression can bias coefficients. Hidden variables can produce apparent nonlinearities. Experimental design should therefore be integrated with equation search. The system can propose measurements that distinguish candidate terms, estimate whether additional precision is worth the cost, and detect when no expression in the current grammar is adequate.

The evaluator for an equation should include more than error and length. It can test numerical stability, units, invariances, parameter sensitivity, causal predictions and compatibility with known limits. A Pareto archive can preserve several expressions that trade simplicity against fit. Human scientists can then inspect a structured frontier rather than one supposedly optimal formula.

Equation discovery also needs an equivalence theory. Two expressions may differ syntactically while representing the same relation after algebraic simplification, a change of coordinates or a rescaling of units. Without canonicalization, a search system can report rediscoveries as diversity; with overly aggressive canonicalization, it can collapse formulations whose domains or singularities differ. The archive should therefore record transformations that preserve meaning, assumptions under which equivalence holds and counterexamples outside those assumptions.

A practical test is extrapolation by regime rather than by random split. A system can fit low-amplitude behaviour and then be tested near a transition, learn one geometry and face another, or be denied a known limiting case. These evaluations reveal whether an expression captured a mechanism or merely interpolated the sampled manifold.

Symmetry can make the search more scientific. Translation, rotation, permutation and conservation symmetries restrict the admissible form of a law and suggest which variables should be grouped. Conversely, a reproducible symmetry violation can signal new physics or a missing interaction. A frontier system would search jointly over expressions, variables, symmetries and experiments. It would ask which measurement most sharply separates rival equations, then update the symbolic archive without discarding alternatives that remain empirically indistinguishable. In that architecture, symbolic regression is not a final curve-fitting step. It is one component in a cycle that compresses evidence, exposes assumptions and proposes discriminating interventions.

A discovered equation is valuable not because it is symbolic, but because its symbols organize interventions, limits and explanations that survive new evidence.

The likely advance is not a universal equation finder. It is a family of domain-grounded systems that combine learned representations, symbolic grammars, physical constraints and active experiments. These systems may expand the range of mechanisms that can be extracted from complex data while making their assumptions unusually explicit.

10.2 AlphaTensor, FunSearch, AlphaProof and AlphaEvolve

Mathematics and algorithms provide some of the clearest demonstrations that machine learning can produce results not explicitly present in its training data. The reason is evaluative strength. A candidate algorithm can be executed and its cost measured. A combinatorial construction can be checked. A formal proof can be verified by a small trusted kernel. These domains allow generators to be adventurous because acceptance does not depend on the generator's own confidence.

AlphaTensor reframed matrix-multiplication algorithm discovery as a game over tensor decompositions and used reinforcement learning to search the space. It found algorithms matching or improving known results for several sizes, including a 47-multiplication construction for a restricted arithmetic setting where two levels of Strassen's method use 49

(Fawzi et al., 2022). The scientific importance was not that a neural network "understood matrices" in the human sense. It was that a formal problem had been converted into a searchable environment with exact feedback.

FunSearch combined a pretrained language model with an evaluator and an evolutionary archive. The model generated program fragments; programs were executed; high-scoring and diverse candidates were retained and used to prompt further generations. The system produced improved constructions in combinatorics, including the cap-set problem, and useful bin-packing heuristics (Romera-Paredes et al., 2024). The central innovation was not free-form language generation but the conversion of language-model proposals into an iterative, executable search process.

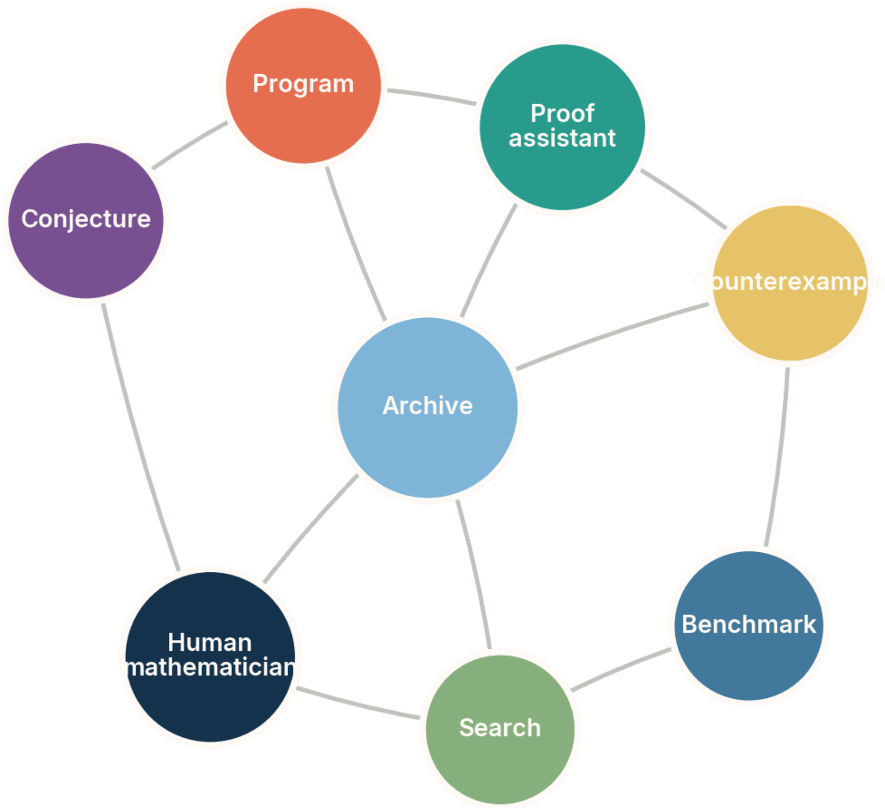


Figure 10.1. Machine-assisted mathematical discovery links conjectures, programs, proof assistants, counterexamples, benchmarks, search, human mathematicians and archives. Conceptual diagram; not an empirical result.

The figure matters because mathematical discovery is broader than proof generation. Programs can reveal patterns; counterexamples refine conjectures; benchmarks measure search; archives preserve constructions; proof assistants certify formal steps; human mathematicians choose definitions and interpret significance. The most productive systems connect these roles. A proof without an intelligible conjecture may be correct but unilluminating, while an empirical pattern without proof remains provisional.

AlphaProof pushed formal theorem proving by combining a language model with reinforcement learning inside the Lean proof environment. The 2025 Nature paper reported silver-medal-level performance on the 2024 International Mathematical Olympiad, with AlphaProof proving three of five non-geometry problems and a geometry system handling another (Hubert et al., 2025). The International Mathematical Olympiad (IMO) acronym is introduced here because it recurs in this chapter. The proofs were formally checkable, but formalization, computation time and domain coverage remained substantial constraints.

AlphaEvolve generalized the generate–evaluate–evolve pattern to code. The 2025 white paper described improvements in data-centre scheduling, hardware circuits and mathematical constructions, including a procedure for multiplying two four-by-four complex matrices using 48 scalar multiplications (Novikov et al., 2025). Independent mathematical work subsequently transformed the construction into a rational-coefficient algorithm over suitable rings (Dumas, Pernet and Sedoglavic, 2025). The result is compelling, but evidence levels should remain clear: AlphaEvolve was initially reported in a system white paper, while later mathematical analysis provides separate validation.

Table 10.2. Why four discovery systems succeeded

System	Key source of reliable progress
AlphaTensor	An exact game formulation and direct verification of tensor decompositions.
FunSearch	Executable programs, scalar evaluation and an evolutionary archive of diverse candidates.
AlphaProof	Reinforcement learning guided by a formal theorem prover with a trusted checking kernel.
AlphaEvolve	Code mutation, automated evaluators and population-based search across many domains.
Symbolic regression systems	Compact executable expressions constrained by fit, simplicity and domain structure.
Human–machine collaboration	Problem formulation, equivalence recognition, interpretation and proof or experiment design.

The table exists to identify the transferable pattern. The systems differ in architecture, but all convert an open-looking intellectual task into a space where many candidates can be rejected cheaply and surviving candidates can be checked independently. The lesson is not that every science should become a game. It is that scientific automation advances when evaluation can be formalized without erasing the phenomenon of interest.

There are important limits. Formal proofs depend on the axioms and definitions supplied. Program search often optimizes a scalar objective that may omit elegance, generality or implementation cost. Search can rediscover known constructions because prior-art comparison is difficult. Compute budgets can be large, and success distributions are uneven. Competition problems are selected for crisp statements and do not represent all mathematical research.

Current frontier work combines large-scale exploration with proof generation and human interpretation. A 2025 preprint on mathematical exploration at scale reported applying AlphaEvolve to dozens of problems, often rediscovering best-known constructions and sometimes improving them, then combining the search with reasoning and proof tools (Georgiev et al., 2025). Such reports are promising but should be read as evolving research rather than settled capability.

The deeper opportunity is interactive mathematics. A system can maintain families of conjectures, generate examples and counterexamples, translate informal claims into formal language, search proofs, and expose the minimal lemmas on which success depends. Human mathematicians can then work with a navigable space of evidence rather than a single answer. This may change which questions are tractable even before it automates the highest-level choice of questions.

CHAPTER 11

11 Biology, Chemistry and Materials

11.1 Prediction, design and mechanistic discovery in biology

Biology illustrates both the power and ambiguity of machine learning for discovery. Prediction can reveal structure in high-dimensional data; generative design can propose sequences and molecules; mechanistic discovery asks why a system behaves as it does. These are distinct achievements. AlphaFold transformed protein-structure prediction from sequence and AlphaFold 3 expanded modelling of biomolecular interactions (Jumper et al., 2021; Abramson et al., 2024). Accurate structure prediction can accelerate science without automatically explaining cellular mechanism or therapeutic effect.

Protein design systems such as ProteinMPNN and RFdiffusion use learned structural regularities to propose sequences and backbones with desired properties (Dauparas et al., 2022; Watson et al., 2023). Protein language models can generate functional sequences across families (Madani et al., 2023). These systems can enter regions not explicitly represented by natural sequences, but novelty must be defined carefully. A sequence can be new as a string while occupying a familiar structural and functional family. Experimental validation determines whether the design folds, binds and works in the relevant context.

Table 11.1. Three levels of biological machine discovery

Level	What would count as success
Prediction	Accurate inference of structure, phenotype or response for held-out biological cases.
Design	A synthesized candidate that achieves a specified property under experimental conditions.
Mechanistic discovery	A causal account that predicts interventions and integrates across contexts.
Ontology formation	A new biological category or state that is stable, measurable and explanatory.
Therapeutic translation	Benefits that survive safety, dosing, population and clinical constraints.
Evolutionary interpretation	A defensible account of how sequence, structure, function and history relate.

The table exists because results are often promoted across levels without sufficient evidence. A binding prediction is not a therapy. A designed molecule is not a mechanism. A cluster in single-cell data is not automatically a cell type. Each transition requires new evaluators and often a different experimental scale. Machine learning is most transformative when it makes these transitions cheaper and more systematic rather than when it relabels prediction as discovery.

Single-cell and spatial omics create another frontier. Models can integrate millions of cells, infer trajectories and identify rare states. Yet batch effects, sampling, tissue preparation and annotation conventions can dominate latent spaces. A biologically new state should persist across donors, protocols and measurement modalities, and ideally support a functional intervention. The representation problem is inseparable from experimental design.

Mechanistic discovery benefits from perturbation data. Clustered regularly interspaced short palindromic repeats (CRISPR) screens, chemical perturbations and genetic interventions create variation that can distinguish pathways. Models can predict perturbation responses and prioritize combinations, but cellular systems contain feedback, redundancy and context dependence. Validation across cell types and organisms is often necessary. The strongest systems link predicted mechanisms to interventions that could falsify them.

Generative models can also search vast chemical and sequence spaces whose size exceeds exhaustive enumeration. Their advantage is learned constraint: they propose candidates more likely to be viable than random combinations. Their risk is distributional conservatism or hidden invalidity. Models may remain close to well-represented families, overestimate properties that are easy to predict, or exploit weaknesses in surrogate assays. Diversity metrics and prospective experiments are essential.

One promising direction is closed-loop design with multiple assays. Instead of optimizing a single predicted property, a system evaluates folding, binding, expression, toxicity, stability and manufacturability at increasing fidelity. Absorption, distribution, metabolism, excretion and toxicity (ADMET) properties are introduced here in full because the acronym is common in drug discovery. Multi-objective archives preserve trade-offs rather than collapsing them into one score.

Another direction is hypothesis-centred multimodal modelling. Literature, sequence, structure, imaging and perturbation data can be linked to a claim graph. The system retrieves the evidence relevant to a proposed mechanism, identifies contradictions and designs an experiment that distinguishes alternatives. This is more ambitious than prediction but more scientifically grounded than unrestricted text generation.

Biological novelty is especially vulnerable to hidden context. A model may discover a pattern specific to one cell line, ancestry group, laboratory protocol or developmental stage and present it as a general mechanism. Cross-dataset validation is necessary but not sufficient because datasets can share preparation biases and annotation conventions. Stronger evidence comes from designed perturbations across environments: change the gene, dose, tissue, donor or assay while preserving a record of what else changed. A mechanism should predict how the effect varies, not only whether an average association repeats.

Clinical translation adds another boundary. Causal relevance in a cell system or model organism may not survive human heterogeneity, treatment history or interacting disease. Machine-learning systems should report the biological population and intervention context for every claim, and treat transfer across those boundaries as a new hypothesis requiring evidence.

This makes closed-loop biology a promising but demanding frontier. Models can select perturbations that maximize information about pathway structure, robotic platforms can execute them, and multimodal assays can measure molecular and phenotypic consequences. Yet the loop needs biological stopping rules. Novelty should not be rewarded when it merely exploits toxicity, stress responses or measurement artefacts. Candidate mechanisms should be tested across scales and compared with simpler explanations. The most valuable automation may therefore be hypothesis triage: identifying the small set of interventions that would most efficiently distinguish competing mechanistic stories, while preserving uncertainty and making negative results reusable.

Biology will probably remain a domain of mixed automation because the evaluator changes with scale. Molecular binding can be measured quickly; organismal function, safety and clinical benefit cannot. The frontier is to automate the lower layers while preserving explicit uncertainty about translation. A system that knows which biological level it has validated is more valuable than one that uses the word “discovery” indiscriminately.

11.2 Chemistry and materials: from screening to verified knowledge

Materials and chemistry offer a natural pipeline from computation to the laboratory. Models screen compositions or structures, generative systems propose candidates, synthesis planners suggest routes, robots execute experiments and instruments characterize products. The pipeline appears ideal for automation because many stages are already digital. The central challenge is that a predicted target, a successful synthesis, a pure phase and a useful material are different claims.

Large-scale materials models have expanded candidate lists dramatically. The GNoME work reported millions of predicted stable structures and hundreds of thousands added to a public database (Merchant et al., 2023). Computational stability is a filtering signal, not proof of synthesizability, novelty or useful properties. The gap between prediction and realization motivates autonomous laboratories.

Figure 11.1. A verified materials result depends on planning, robotics, instruments, analysis, modelling, safety and provenance around a declared target. Conceptual diagram; not an empirical result.

The figure matters because a laboratory result cannot be attributed to the model alone. Planning selects conditions, robotics executes them, instruments produce signals, analysis assigns phases or properties, safety constrains actions and provenance makes the claim reconstructable. A failure or success at any node changes the interpretation. The central provenance node is deliberate: without a traceable path from target to measurement, rapid experimentation produces data but not cumulative evidence.

A-Lab, an autonomous laboratory for accelerated solid-state materials synthesis, is a valuable case because it integrated literature-derived recipes, thermodynamic calculations, active learning, robotics and X-ray diffraction (XRD) analysis. Over seventeen days, the system realized 36 compounds from 57 targets (Szymanski et al., 2023). An Author Correction published in January 2026 revised parts of the reporting and underscores a general lesson: autonomous discovery claims require the same, or stronger, scrutiny as human-led work (Szymanski et al., 2026).

The correction should not be used to dismiss autonomous laboratories. It should improve their standards. Systems need transparent target definitions, phase-identification uncertainty, raw data access, independent analysis and clear distinctions among predicted novelty, synthesis success and purified product. Rapid automated campaigns magnify both productivity and systematic error. Corrections are evidence that institutional verification remains part of the discovery architecture.

Table 11.2. Stages at which a computational materials "discovery" can fail

Stage	Typical unresolved question
Database novelty	Is the candidate absent because it is new, differently represented or missing from the database?
Predicted stability	Does the computational model cover the relevant phases, temperature and kinetics?
Synthesis planning	Are the proposed precursors, conditions and route physically feasible and safe?
Experimental realization	Was the target phase produced, in what purity and with what reproducibility?
Property validation	Does the measured material exhibit the intended performance under realistic conditions?
Scientific significance	Does the result reveal a new mechanism, capability or design principle rather than one more candidate?

The table exists because every stage changes the semantics of "discovered." A candidate absent from a database is a database novelty. A stable calculation is a prediction. A diffraction match is evidence about phase identity. A useful property is an engineering result. A mechanism requires additional experiments. Reporting the stage precisely makes progress comparable and prevents later corrections from appearing to reverse more than they actually do.

Frontier laboratories are moving beyond scalar optimization toward principle discovery. Real-time experiment–theory loops can select measurements that discriminate models, while microfluidic platforms explore synthesis and mechanism with low material cost. Multi-agent systems separate planning, control and analysis. Autonomy frameworks assess not only throughput but adaptation, decision scope and human oversight. The research opportunity is to make laboratories produce explanatory models, not merely optimized samples.

Surrogate models remain a weak link. A property predictor can be highly accurate near known materials and fail in a new chemical regime. Multi-fidelity methods combine calculation, proxy assays and definitive measurements. Uncertainty should determine which fidelity is used. Novel regions deserve stronger physical checks precisely because the model has less support there.

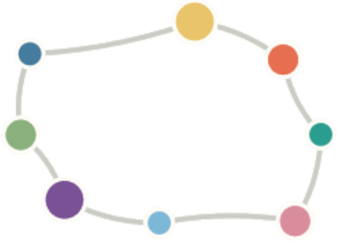
Closed-loop chemistry also raises safety and environmental questions. A planner that searches reaction space may propose unstable combinations or generate hazardous waste. Constraint systems, reaction filters, physical interlocks and human approval should be integrated before optimization. Safety data and failed reactions should enter the archive, not disappear because they lower performance metrics.

The most interesting future result will not be the largest number of autonomous experiments. It will be a system that identifies a previously unrecognized organizing principle, selects the decisive experiments, survives independent replication and leaves an auditable record from proposal to mechanism. That is where automation becomes scientific discovery rather than accelerated screening.

PART IV

Part IV. Open-ended Science and the Automation Frontier

The automation frontier is an institutional architecture, not a single model score.



CHAPTER 12

12 Search Without a Fixed Destination

12.1 Novelty search, quality-diversity and the value of stepping stones

Most optimization begins with an objective. Yet a distant objective can create a deceptive landscape: the behaviours that eventually enable success may initially score poorly. Novelty search replaces direct objective maximization with a reward for behavioural difference (Lehman and Stanley, 2011). Quality-diversity (QD) methods preserve many high-performing solutions across behavioural niches. Their premise is that diversity is not merely a side effect or fairness constraint. It is a computational resource that preserves stepping stones.

The distinction between genotype, phenotype and behaviour is important. Two programs may differ syntactically while behaving identically. A useful novelty metric should operate in a space connected to function, strategy or mechanism. Defining that space is difficult because human designers can predetermine which differences count. Learned behaviour descriptors and adaptive niches attempt to loosen this constraint, but they can drift toward features that are easy to vary rather than scientifically meaningful.

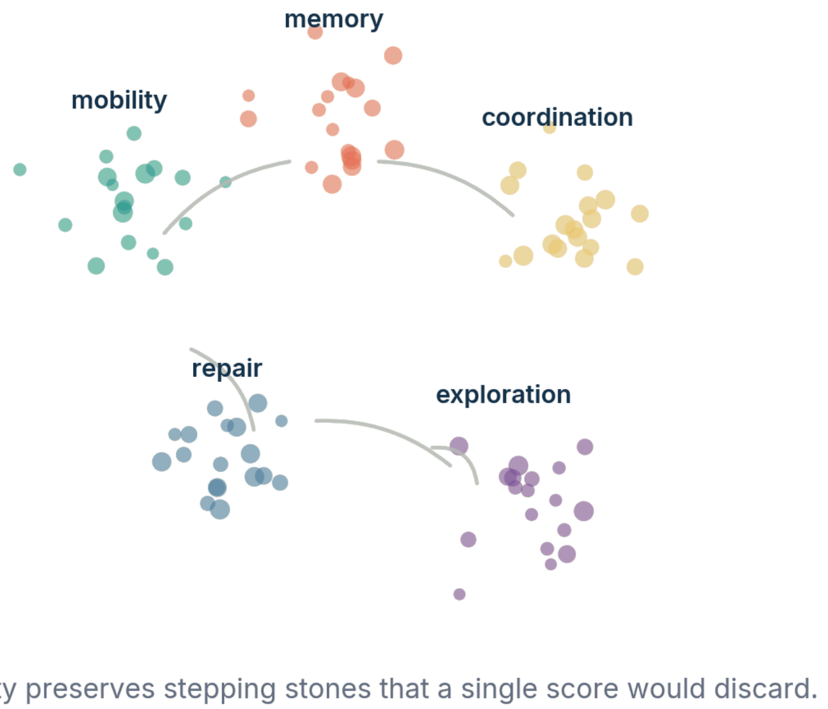


Figure 12.1. An open-ended archive preserves diverse behavioural regions and the bridges that later become stepping stones. Conceptual diagram; not an empirical result.

The figure matters because the archive is not a leaderboard. Each cluster contains variations that may be mediocre under a global score but valuable within a niche. The thin bridges represent transfers and recombinations that objective-only search might never preserve. In science, an abandoned method, anomalous dataset or partial theory can play the same role. Open-ended systems need memory for possibilities that are not currently fashionable.

MAP-Elites, short for Multi-dimensional Archive of Phenotypic Elites, discretizes a behavioural space and stores the best candidate found in each cell (Mouret and Clune, 2015). It produces a repertoire rather than one optimum. The method has been used in robotics, design and procedural generation. Its limitation is that the axes are normally chosen in advance. The archive can become rich within an impoverished map. Recent work explores learned descriptors, hierarchical archives and illumination across changing tasks.

Table 12.1. Objective-driven and open-ended search ask different questions

Search regime	Question it asks
Single-objective optimization	Which candidate maximizes one declared score?

Search regime	Question it asks
Multi-objective optimization	Which candidates form the best trade-off frontier among several scores?
Novelty search	Which behaviours are most different from those already encountered?
Quality-diversity search	Which high-quality candidate occupies each behavioural niche?
Coevolution	Which challenges and solutions continue to improve against one another?
Open-ended discovery	Which mechanisms can continually create new niches, tasks and representations?

The table exists because open-endedness is often mistaken for adding randomness to optimization. Novelty search still depends on a behaviour metric. Quality-diversity still includes quality. Coevolution can cycle or collapse. Genuine open-endedness requires the capacity to create new dimensions of variation, not merely to fill a fixed grid. The table marks a sequence of increasingly demanding questions rather than a claim that one regime is always superior.

Paired Open-Ended Trailblazer (POET) coevolves environments and agents, transferring solutions among environments so that challenges can grow while remaining solvable (Wang et al., 2019). This demonstrates how tasks themselves can become search objects. A hard environment that is impossible from scratch may become reachable through a curriculum of stepping stones. Similar principles can apply to scientific agents that generate benchmark problems, counterexamples or simulated worlds for one another.

Open-ended search creates evaluation problems. If the system changes the tasks, behaviours and representations, a fixed metric cannot certify progress. Researchers can measure diversity, transfer, complexity, novelty persistence and the emergence of qualitatively new capabilities, but these remain partial. Human interpretation returns because significance is not fully specified in advance. Open-endedness can produce endless variation without useful discovery.

Safety also changes. A population of diverse agents or experiments explores behaviours that designers did not enumerate. Constraint systems, sandboxing and tiered access become necessary. Archives should record lineages and rejected candidates, and experiments should remain inside a safety envelope even when objectives evolve. Open-endedness without containment is not a research programme; it is uncontrolled variation.

The scientific opportunity is selective open-endedness. Systems can explore broadly in simulation or formal spaces, preserve diverse hypotheses and propose new behavioural dimensions, while stronger evaluators and humans decide what enters the physical world or scientific record. This architecture accepts that interesting discoveries may require detours without granting every detour equal authority.

12.2 Self-generated tasks and self-improving agents

A system becomes self-improving when some of the objects it modifies are components of its own problem-solving process: prompts, tools, code, memory, search policies or evaluators. The strongest popular image is recursive self-improvement without external guidance. Current systems are narrower. They typically operate inside a fixed benchmark, use a frozen foundation model, edit an agent scaffold or codebase, and accept changes only when empirical tests improve. This is still important, but it should not be confused with unrestricted improvement of intelligence.

The Darwin Gödel Machine (DGM), published at the 2026 International Conference on Learning Representations, evolves coding agents that modify their own code and are empirically evaluated on software benchmarks (Zhang et al., 2026). The reported system improved from 20.0 to 50.0 per cent on its selected Software Engineering benchmark (SWE-bench) tasks and from 14.2 to 30.7 per cent on the full Polyglot benchmark (Jimenez et al., 2024). An archive of diverse agents enabled branching lineages rather than a single hill-climbing path. The result demonstrates open-ended scaffold improvement under a fixed external evaluator.

Table 12.2. What is and is not self-improving in current agent systems

Component	Typical status in present research
Foundation model weights	Usually fixed during the self-improvement run.
Agent code and prompts	Editable through generated patches, search and benchmark feedback.
Tool repertoire	Sometimes expanded or reconfigured within a permitted interface.
Evaluator and benchmark	Usually fixed externally, which prevents circular acceptance but limits scope.
Task distribution	Occasionally generated or coevolved, though still constrained by a framework.
Scientific values and safety constraints	Remain human-defined and should not be silently optimized away.

The table exists to prevent "self-improving" from functioning as a magical adjective. Current systems improve selected components while relying on fixed external foundations. That externality is a strength because it prevents the system from declaring its own changes successful under an altered standard. It is also a limitation because progress may overfit the benchmark. The real research problem is how to expand the editable boundary without losing independent evaluation.

Self-generated tasks can reduce benchmark saturation. An agent proposes challenges just beyond current competence, while another agent or formal environment verifies them. This resembles automatic curriculum learning. The difficulty is task validity. Generated problems may be ambiguous, trivial, impossible or contaminated by hidden assumptions. Formal domains provide a strong filter; open scientific questions do not. Task generators need novelty checks, solvability estimates and mechanisms for identifying whether a new task teaches a reusable capability.

Self-modification creates a credit-assignment problem. A code change can improve one benchmark by accident, harm unmeasured capabilities or increase cost. Population-based testing and regression suites reduce risk. Behavioural archives preserve alternatives if a dominant lineage later fails. Versioned provenance makes each improvement reversible. These are ordinary software-engineering practices elevated into the epistemic core of self-improvement.

The evaluator can itself improve, but doing so introduces circularity. If the same system changes the test and the solution, progress becomes difficult to interpret. One response is constitutional separation: generators can propose evaluator changes, but acceptance requires external review, hidden tests or formal meta-properties. Another is evaluator diversity: several independently trained or implemented systems must agree that the new test is harder without merely being different.

Recursive improvement also faces diminishing returns and resource constraints. Easy scaffold optimizations are discovered first. Later changes require more compute, longer evaluation and broader tests. A system can spend increasing resources to obtain benchmark-specific gains. Claims of continuing takeoff therefore require evidence about transfer, marginal cost and the emergence of new capability classes, not only a rising score.

The frontier is empirical self-improvement with controlled boundaries. Systems can search their own tools, representations and workflows while immutable logs, external evaluators and safety constraints preserve accountability. Open-ended archives reduce premature convergence. Human researchers decide when a newly discovered modification changes the scientific question rather than merely the implementation.

> *Self-improvement is scientifically meaningful when the system can show not only that it changed, but which independent evidence establishes that the change broadened reliable competence.*

This field may eventually produce agents that invent new learning algorithms or discovery procedures. The path is not to remove evaluation but to make evaluation more layered, adaptive and difficult to corrupt. The self-improving system remains part of a larger scientific institution that it does not fully control.

CHAPTER 13

13 The Artificial-Intelligence Scientist

13.1 From research assistant to partially autonomous research system

An artificial-intelligence research agent combines a large language model (LLM), tools, memory and workflow control to perform parts of scientific work. The phrase “AI scientist” is rhetorically powerful but technically broad. Systems may search literature, generate hypotheses, write code, run computational experiments, analyse data, draft manuscripts or plan wet-laboratory validation. Their autonomy is bounded by tools, datasets, permissions and the scientific domain. The important question is not whether the label applies, but which epistemic functions are automated and which evaluators remain external.

The AI Scientist system reported in Nature in 2026 automated a loop of idea generation, coding, experiments, visualization, manuscript writing and automated review within machine-learning research (Lu et al., 2026). One generated paper passed an initial workshop review stage. This is a meaningful end-to-end demonstration, but the domain had executable experiments and a comparatively structured publication format. The result does not establish general competence at selecting important scientific questions or producing reliable knowledge across disciplines.

Co-Scientist, also published in Nature in 2026, used a multi-agent architecture for hypothesis generation, reflection, ranking, evolution and synthesis, with scientists in the loop and experimental validation in biomedical settings (Gottweis et al., 2026). Robin, a separate multi-agent system, automated hypothesis generation and data analysis for experimental biology and reported a closed discovery workflow (Ghareeb et al., 2026). These systems show that orchestration can turn general models into useful scientific collaborators when paired with domain tools and human validation.

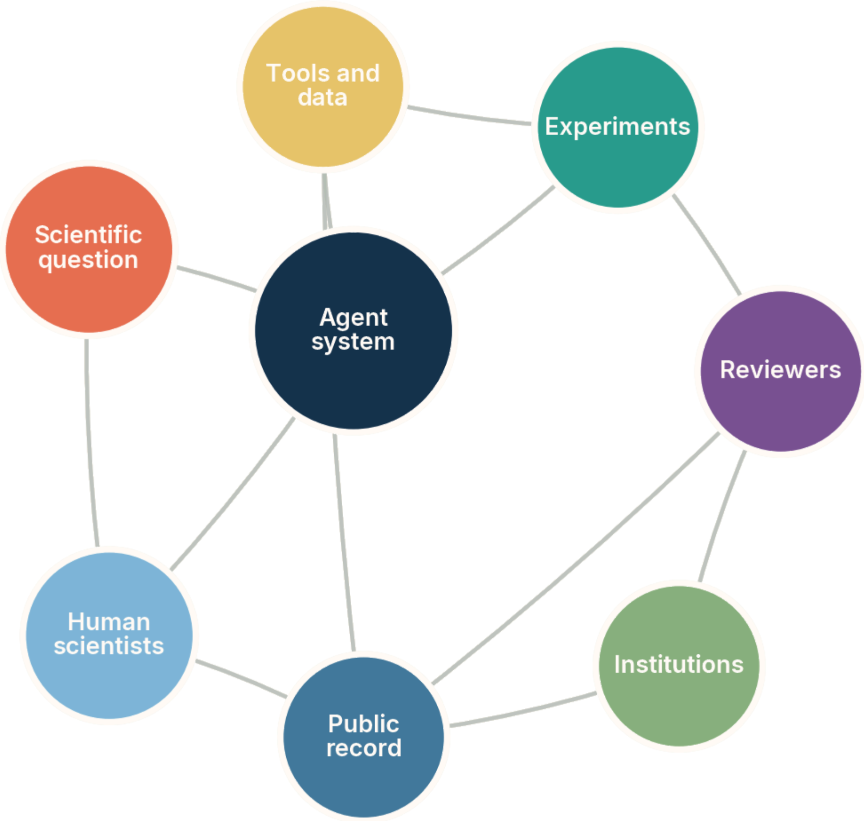


Figure 13.1. An artificial-intelligence research agent is one node in a scientific institution that includes tools, experiments, reviewers, public records and human scientists. Conceptual diagram; not an empirical result.

The figure matters because the phrase “AI scientist” can make the agent appear self-sufficient. In reality, the agent depends on literature infrastructure, experimental access, reviewer standards, institutional permissions and human interpretation. The public record provides memory beyond the model. Reviewers and experiments create independent resistance. The system is most reliable when these nodes remain visible rather than being compressed into an opaque claim of autonomy.

Other 2026 systems specialize in parts of the workflow. Empirical Research Assistance (ERA) was reported as an agentic system that helps scientists write expert-level empirical software, combining domain-specific language design, code search and evaluation (Aygün et al., 2026). OpenScholar retrieves relevant passages from tens of millions of open-access papers and synthesizes citation-backed answers (Asai et al., 2026). Agentic X-ray systems and autonomous microscopy platforms connect language-mediated planning to instruments. The trend is modular scientific infrastructure rather than one universal scientist.

Table 13.1. Current scientific-agent capabilities and their strongest available evaluator

Automated function	Most credible form of evaluation
Literature synthesis	Citation entailment, retrieval recall and expert comparison against the relevant corpus.
Hypothesis generation	Novelty review, mechanistic coherence and prospective discriminating experiments.
Code and analysis	Execution, tests, reproducibility and comparison with expert pipelines.
Experiment planning	Safety review, feasibility checks and expected discrimination among hypotheses.
Instrument control	Physical constraints, calibration, recovery from faults and traceable action logs.
Manuscript production	Claim–evidence alignment, independent peer review and replication rather than prose quality.

The table exists to connect each capability to an evaluator that lies outside linguistic fluency. A coherent literature review can still omit decisive papers. An original hypothesis can be untestable. Executable code can implement the wrong analysis. A polished manuscript can overstate its evidence. Scientific-agent evaluation should therefore be function-specific and prospective whenever possible.

Agent scaffolds add planning, memory, role separation and tool use. They can substantially improve workflow execution, but they do not automatically install scientific norms. Multiple agents based on the same model can agree because they share priors and errors. Reflection can elaborate an early mistake. Retrieval can provide evidence that the agent fails to use. The scaffold is a computational institution, and like human institutions it requires checks against conformity and self-confirmation.

The current frontier is mixed-initiative science. The agent proposes and executes within explicit boundaries; humans set or revise high-level questions, approve consequential experiments, inspect disagreements and interpret significance. Over time, more functions can move into the automated layer as evaluators strengthen. This staged view is more useful than debating whether a system is “really” a scientist.

The deepest opportunity may be accessibility. Scientific agents can make advanced tools, code and literature available to smaller laboratories and interdisciplinary teams. Yet centralization can also narrow science if many researchers rely on the same models, corpora and evaluators. Capability and epistemic diversity must be designed together.

13.2 Scientific reasoning, failure modes and institutional verification

A workflow can close without the reasoning being scientific. Scientific reasoning includes responsiveness to evidence, active search for disconfirmation, comparison of rival explanations, replication and revision of confidence. An agent may execute every step of a template while anchoring on an early hypothesis or ignoring contradictory results. Outcome-only evaluation can miss this because a correct final answer may have been reached through an unreliable process.

A 2026 preprint evaluated large-language-model scientific agents in more than 25,000 runs across eight domains and analysed both outcomes and reasoning traces (Ríos-García et al., 2026). The authors reported that the base model explained far more variance than the agent scaffold, that relevant evidence was frequently ignored, and that refutation-driven belief revision was uncommon. The study is a preprint and its operational definitions should be scrutinized, but it provides an important warning: better orchestration alone may not create epistemic discipline.

Table 13.2. Epistemic behaviours that a scientific agent should be tested for

Behaviour	Observable test
Evidence uptake	Does new evidence change ranking, confidence or the next action in the predicted direction?
Refutation seeking	Does the system select experiments that could falsify its preferred hypothesis?
Alternative preservation	Does it maintain rival explanations instead of collapsing too early?
Belief revision	Can it reverse a conclusion when decisive contradictory evidence appears?
Multi-test convergence	Does it seek independent measurements rather than repeated variants of one test?
Provenance fidelity	Can every major claim be traced to data, code, source and decision history?

The table exists because scientific quality cannot be inferred from the final report alone. These behaviours can be tested experimentally by presenting controlled contradictions, hidden confounders, ambiguous results and opportunities for decisive tests. A system that succeeds only when evidence agrees with its first hypothesis is not a reliable discoverer. Process evaluation should complement outcome evaluation, not replace it.

Automated peer review has similar limits. Models can identify missing citations, statistical inconsistencies and unclear writing, but they can share the author model's assumptions or favour conventional styles. The AI Scientist study used automated reviewing as part of its loop, which increases throughput but also risks correlated acceptance. Independent human review, hidden evaluations and replication become more important as production costs fall.

The volume problem is institutional. If agents can generate thousands of plausible papers, review systems can be flooded and negative externalities shifted to human readers. Nature Communications authors have argued for safeguarding frameworks that combine human regulation, agent alignment and environmental feedback (Tang et al., 2025). Practical controls include rate limits, provenance requirements, disclosure of automated contributions, evidence thresholds and responsibility assigned to human or institutional sponsors.

Scientific monoculture is another risk. A Nature study of tens of millions of papers reported that artificial-intelligence tools were associated with higher individual productivity and impact but a contraction in the collective topical focus of science (Hao et al., 2026). The causal interpretation of observational bibliometrics requires caution, yet the mechanism is plausible: shared tools make data-rich, benchmarkable questions easier and steer attention toward similar regions. Automated discovery should therefore measure portfolio diversity, not only productivity.

Security and integrity matter because agents act through tools and external corpora. Prompt injection, poisoned retrieval, manipulated databases and unsafe code can redirect research. Literature agents need source trust and citation checks. Laboratory agents need permission boundaries and physical interlocks. Code agents need sandboxes and dependency control. A scientific record should distinguish untrusted input from reviewed evidence.

Institutional verification can be computationally strengthened. Claims can be submitted with executable environments, machine-readable provenance, predefined analysis plans and links to raw data. Review agents can reproduce calculations and generate adversarial tests. Human reviewers can focus on conceptual novelty, external validity and significance. Replication marketplaces or coordinated laboratories can test high-value claims prospectively.

The frontier is not maximum autonomy. It is maximum justified delegation. A scientific function can be delegated when the system's competence is measured, its failures are detectable, its actions are bounded and its outputs enter an independent verification process. This principle supports ambitious automation while preserving the self-correcting structure that gives science its authority.

CHAPTER 14

14 Structural Limits and Movable Boundaries

14.1 Limits that scaling cannot erase

Some failures diminish with better models, more data and more computation. Others follow from the structure of the problem. A comprehensive frontier map must separate them. Otherwise every impossibility is dismissed as temporary and every engineering weakness is treated as metaphysical. Structural limits do not imply that useful work is impossible. They identify where the claim, data or interaction must change.

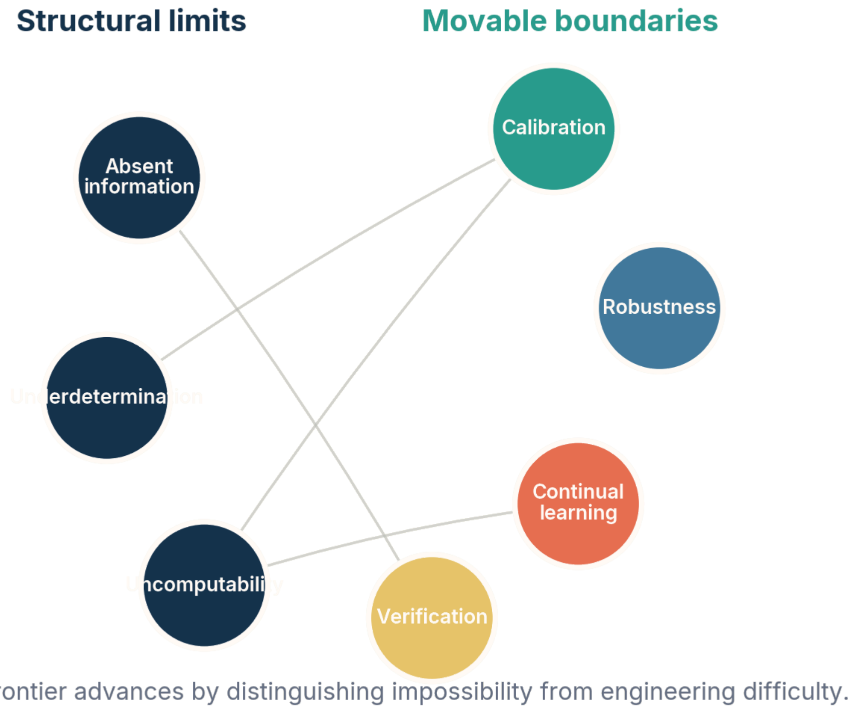


Figure 14.1. Structural limits and movable boundaries interact, but they require different research responses. Conceptual diagram; not an empirical result.

The figure matters because structural and movable constraints are not isolated. Better calibration can reveal when absent information matters, but it cannot create the information. Verification can certify a formal claim, but not global historical priority. Continual learning can update a model, but it cannot guarantee that a finite record determines the correct future. Progress comes from moving the engineering boundary while respecting the remaining structure.

The first structural limit is absent information. A system cannot infer a distinction that leaves no observable trace in its data, sensors, tools or interactions. It can use priors to guess, but a guarantee would require evidence. The remedy is to obtain a new measurement, intervention or source. This principle applies equally to hidden biological variables, private historical records and adversarial strategies designed to imitate normal data.

The second limit is underdetermination. Finite evidence can support several models that diverge outside the observed region. More data can reduce the set, but no finite dataset eliminates all logically possible continuations without assumptions. Inductive bias is unavoidable. The scientific response is to expose assumptions, preserve alternatives and seek observations that make them disagree.

The third limit is identification. Observationally equivalent causal structures, non-identifiable latent variables and duplicated representations cannot be separated under the current design. Additional assumptions or interventions can make a narrower problem identifiable. Scaling a predictor on the same observations does not change the equivalence class.

Table 14.1. Structural limits and the appropriate response

Structural limit	What a responsible system should do
Absent information	State that the question is not answerable from current evidence and request a new channel.
Underdetermination	Maintain alternatives and disclose the inductive assumptions selecting among them.
Non-identifiability	Propose interventions or restrict the claim to identifiable quantities.
Undecidability or intractability	Offer bounded heuristics, certificates for solved instances and explicit coverage.
Incomplete prior-art record	Report a scoped search and probabilistic priority claim, not absolute novelty.
Specification gap	Treat success on the proxy as provisional and search for counterexamples or external validation.

The table exists to translate limits into behaviour. A system need not respond to impossibility with silence. It can narrow a question, identify missing evidence, return a set rather than a point, or provide a certificate for a solved instance. The irresponsible response is unjustified confidence. The productive response turns the limit into a plan for changing the information structure.

Computational limits include undecidable problems and intractable search. Learning can exploit distributions of practical instances and generate useful heuristics. It can also produce certificates, such as formal proofs, that are cheap to check even if hard to find. The claim must remain distributional or instance-specific. Strong empirical performance does not imply a universal decision procedure.

Historical novelty has a special limit. No accessible corpus contains every unpublished idea, language, patent, notebook and oral tradition. Semantic equivalence is itself imperfect. A system can provide a reproducible prior-art search and estimate that no close precedent was found. It cannot establish absolute firstness from an incomplete record. Human scholarship lives with the same limit; machines should not conceal it.

Specification gaps are also structural relative to an objective. An optimizer cannot infer an intention that is neither encoded nor observable in feedback. Inverse learning can infer preferences from behaviour, but behaviour underdetermines values and can be inconsistent. Dialogue, oversight and corrigibility can improve alignment, yet the system cannot guarantee fidelity to an intention that the institution itself has not resolved.

These limits define an ethics of claims. Models should distinguish prediction from identification, evidence from assumption, local novelty from historical priority, and heuristic success from guarantee. This does not diminish machine learning. It makes its achievements comparable and allows research to focus on boundaries that can genuinely move.

14.2 Boundaries with plausible paths forward

Many serious limitations are not structural. They persist because current representations, evaluators, data systems or institutions are weak. Calibration under shift, compositional generalization, continual learning, causal representation, mechanistic interpretability, long-horizon planning and autonomous experimentation can all improve substantially. None is guaranteed to be solved by scale alone, but each has identifiable research levers.

Table 14.2. Movable boundaries and the research leverage that may shift them

Movable boundary	Promising leverage
Uncertainty under shift	Living calibration, conformal adaptation, explicit shift models and abstention policies.
Compositional generalization	Modular architectures, program induction, meta-learning and tests of systematic recombination.
Continual learning	Layered memory, sparse updates, replay, versioned knowledge and longitudinal evaluation.

Movable boundary	Promising leverage
Causal representation	Interventions, multiple environments, identifiable latent models and physical testbeds.
World-model reliability	Object persistence, conservative rollouts, uncertainty and frequent real-world correction.
Scientific-agent reliability	Process evaluation, independent verifiers, provenance and bounded tool permissions.

The table exists to pair ambition with an experimental programme. "Better reasoning" is too vague. Compositionality can be tested on recombinations designed to defeat memorized templates. Continual learning can be measured over recurring tasks and revisions. Causal representations can be evaluated against known interventions. Scientific agents can be tested for belief revision and evidence use. Clear leverage creates falsifiable progress.

A boundary should be considered moved only when improvement survives prospective, cross-domain and adversarial evaluation. A higher benchmark score can reflect better fitting to the test ecology rather than a broader capability. Frontier claims should therefore report the operating envelope: distributions, tools, time horizon, intervention rights, compute budget and acceptable failure rate. Progress is more convincing when the system transfers to a new institution, recognizes when its assumptions fail and benefits from correction without losing prior competence. These criteria turn vague optimism into measurable boundary movement.

Research portfolios should compare interventions, not only models. Better data collection, a stronger sensor, a formal checker or a changed institutional workflow may move a boundary more than another order of magnitude of compute. Recording cost and marginal information gain helps direct effort toward the actual bottleneck.

Compositional generalization has already moved. Early sequence models failed dramatically when familiar primitives were recombined in unfamiliar ways. Meta-learning and training curricula have produced more human-like systematic generalization on controlled tasks (Lake and Baroni, 2023). The open question is transfer to the complexity and ambiguity of natural language, science and embodied action. Progress will require benchmarks that separate true recombination from pattern overlap.

Continual learning is likely to advance through architectural separation. External memory and retrieval handle rapidly changing facts; adapters or modules localize domain updates; slower consolidation changes shared representations. Mechanistic tools may identify where concepts are stored, while versioned knowledge graphs preserve provenance. The boundary will move when systems can revise connected beliefs coherently rather than patch isolated associations.

Causal representation learning has a particularly clear theoretical frontier. Recent work states identifiability conditions under interventions, environment changes and structured latent models. Physical testbeds can determine whether recovered variables correspond to real controllable factors. The challenge is to scale these guarantees from synthetic systems to biology, climate and social processes without hiding crucial assumptions.

World models may become more useful through interaction. Persistent object representations, latent state estimation and action-conditioned prediction can support planning. The key evaluation is counterfactual accuracy under interventions, not visual realism. Systems should expose uncertainty and permit model ensembles to disagree. Real-world corrections should update the model before planning errors compound.

Mechanistic interpretability may shift from descriptive feature naming to causal engineering. Researchers can intervene on circuits, trace information flow and test whether mechanisms transfer across inputs. Sparse representations can make models easier to inspect, but the field needs standardized faithfulness tests. An explanation should predict the effect of a controlled internal change.

Scientific agents will improve when epistemic behaviour becomes a training target. Models can be rewarded for seeking falsification, preserving alternatives, updating on evidence and citing support accurately. Simulated research environments with known ground truth can provide dense training, while real scientific use supplies prospective evaluation. Scaffold design will still matter, but it cannot compensate indefinitely for base models that do not use evidence reliably.

The common pattern is stronger coupling between learning and correction. Movable boundaries shift when systems receive feedback that identifies the mechanism of failure, not merely a lower score. Rich evaluators, interventions, provenance and modularity make that feedback possible. The next generation of machine learning may be defined less by larger static models than by better organized cycles of criticism and revision.

CHAPTER 15

15 Where the Most Interesting Novelty May Be

15.1 Verification-first scientific infrastructure

The largest opportunity in machine discovery may not be a model that generates more hypotheses. It may be infrastructure that makes hypotheses easier to test, compare and accumulate. Science already produces more papers, data and code than any individual can evaluate. Artificial intelligence can increase output further. Without stronger verification, the result may be noise, duplicated work and confident error. Verification-first infrastructure treats evidence flow as the primary object of automation.

Such infrastructure begins with literature that is computationally traceable. Retrieval systems should identify passages, not merely documents, and distinguish direct evidence from commentary. Claims should link to datasets, code, assumptions and related contradictions. OpenScholar demonstrates the scale possible for citation-backed synthesis across millions of papers (Asai et al., 2026). The next step is claim-level entailment, provenance and update when a source is corrected or retracted.

Figure 15.1. The discovery frontier is a constellation of better questions, world models, causal experiments, formal verification, open-ended search, scientific memory, living evaluation and collective intelligence. Conceptual diagram; not an empirical result.

The figure matters because no single breakthrough is sufficient. Better world models without experiments can become elaborate simulators. Open-ended search without verification produces variation. Formal methods without better questions optimize narrow specifications. Scientific memory without living evaluation accumulates stale claims. The central infrastructure node represents interfaces that allow these capabilities to correct one another.

Executable papers are one component. Code, environment, data transformations and figures can be rerun automatically. Agents can test whether results survive dependency changes, alternate seeds and reasonable analysis choices. Formal schemas can declare primary outcomes, exclusions and stopping rules. This does not automate conceptual validity, but it makes hidden analytic flexibility visible and reduces the cost of replication.

Claim graphs are another component. A paper contains many claims with different evidence. Representing them as linked objects allows a correction to propagate to dependent conclusions. Competing hypotheses can share evidence without being merged. Agents can identify which new experiment would resolve the largest disputed region. The graph becomes a research map rather than a static library.

Table 15.1. Infrastructure that may yield more discovery than a larger generator alone

Infrastructure	Scientific leverage
Claim-level literature graph	Connects assertions to evidence, contradiction, correction and priority.
Executable research object	Makes code, data and analysis reproducible and testable by agents.
Formal and empirical verifier services	Provide shared checking for proofs, programs, models and measurements.
Versioned scientific memory	Preserves revisions, negative results and alternative hypotheses.
Federated laboratory network	Routes high-value candidates to independent experimental validation.
Living benchmark and incident archive	Updates evaluation as systems, users and environments change.

The table exists because these resources create positive externalities. A better generator helps its owner; a shared verifier or claim graph can improve an entire field. Infrastructure also changes incentives by making reproducibility, negative results and corrections more visible. It may reduce the advantage of producing many polished claims and increase the value of producing evidence that others can reuse.

Federated laboratories could provide prospective validation. Agents rank hypotheses by expected value and uncertainty; independent sites volunteer capacity; protocols and raw data are standardized; results are aggregated with site effects modelled explicitly. This would not remove human expertise. It would make replication a routinized service rather than an exceptional act.

Formalization tools can expand the region of cheap verification. Natural-language claims can be translated into testable specifications, with humans approving the translation. Program properties, mathematical lemmas, data constraints and experimental protocols can then be checked automatically. Autoformalization remains error-prone, so the mapping between informal intent and formal object must stay visible.

The most valuable novelty may appear in evaluator design itself. A new assay, proof representation, benchmark environment or measurement instrument can unlock discoveries that generators could not previously validate. Historically, scientific instruments changed the space of questions. Computational verifiers and autonomous experiment networks may play a similar role.

Verification-first infrastructure is slower to dramatize than a model demo. Its payoff is cumulative. It lowers the cost of being wrong, makes disagreement productive and allows more ambitious generators to operate safely. The scientific frontier advances when novelty can encounter resistance quickly.

15.2 Human–machine collective discovery and a frontier research agenda

The future of machine discovery is likely to be collective rather than solitary. Humans, models, instruments, databases and institutions contribute different forms of search and judgment. Machines offer scale, memory, simulation and rapid iteration. Humans contribute shifting values, embodied context, conceptual reframing and responsibility. The objective is not to assign every function permanently, but to design interfaces through which strengths combine without making failures invisible.

Collective intelligence requires diversity. If every laboratory uses the same foundation model, retrieval index and evaluator, individual productivity may rise while the scientific portfolio narrows. Hao and colleagues' 2026 analysis reported this tension at scale: artificial-intelligence use was associated with greater individual impact and productivity but a contraction in collective focus. A system-level research agenda should therefore measure topic diversity, methodological diversity, institutional access and concentration of epistemic infrastructure.

Table 15.2. A frontier research agenda for machine-assisted discovery

Research direction	Decisive evidence of progress
Unknown-aware learning	Reliable abstention and escalation across predeclared and newly encountered shifts.
Causal concept formation	Latent variables that support intervention and transfer in real physical testbeds.
Open-ended search	Sustained creation of useful new niches without collapse into trivial variation.
Scientific process learning	Improved evidence uptake, falsification seeking and belief revision, not only final scores.
Verifier construction	Cheaper independent checks that correlate prospectively with real-world validity.
Collective diversity	Higher productivity without contraction of questions, methods or participating institutions.

The table exists to make the agenda falsifiable. Each direction has a failure mode that ordinary benchmarks may hide. Unknown-aware learning must be tested on shifts not used to tune the detector. Causal concepts must survive intervention. Open-ended search must generate useful diversity rather than decorative variation. Scientific process learning must change how evidence affects action. Collective diversity must be measured at the field level.

One promising interface is disagreement routing. Several models, simulations or experts analyse a claim. Agreement on routine cases enables automation. Structured disagreement triggers stronger evidence or broader review. The system does not treat consensus as truth; it uses disagreement as a signal about model sensitivity and missing assumptions. Diversity must be substantive, based on different data, methods or perspectives, not merely different role prompts applied to the same model.

Another interface is question markets or portfolios. Agents generate candidate questions and estimate tractability, significance, cost and neglectedness. Humans and institutions allocate resources across a diverse portfolio rather than choosing only the highest predicted return. The mechanism can reserve capacity for high-uncertainty questions, replication and tool building. This protects science from becoming a narrow optimization process.

Education and access are part of the frontier. Scientific agents can help researchers use advanced statistics, formal methods and laboratory automation. They can translate across disciplines and languages. But dependence on proprietary infrastructure can exclude institutions and reduce auditability. Open standards, shared benchmarks, public verifiers and local deployment options will influence which communities can participate in machine-assisted discovery.

Governance should follow capability at the level of action. Literature synthesis needs citation and privacy standards. Code execution needs sandboxing. Laboratory control needs physical safety and permissions. Self-improvement needs regression tests and immutable logs. Publication agents need disclosure and accountability. Broad principles become effective only when attached to specific interfaces and evidence records.

Research should also study failure ecology. When many agents interact, errors can spread through citations, generated datasets and shared evaluators. A false but plausible claim can be repeated until retrieval systems treat it as consensus. Provenance, source diversity and correction propagation are therefore central capabilities. Scientific memory must be able to forget authority while preserving history.

The most interesting novelty may be institutional rather than algorithmic: new forms of peer review, distributed replication, machine-readable argument and dynamic allocation of experimental capacity. These can change what questions are affordable and whose ideas receive evaluation. Machine learning becomes an engine of science when it reorganizes the cost structure of criticism, not only the cost of generation.

The ultimate frontier is a discovery system that can surprise us, show us why the surprise matters, design the test that could defeat it, and preserve the entire path for others to challenge.

No present system satisfies that description generally. Many components already exist in narrow form. The research programme is to connect them without erasing their boundaries: generators that remain separate from judges, world models corrected by experiments, memories that preserve contradiction, agents constrained by institutions and human communities capable of redefining significance. This is a realistic path toward extending both science and the range of work computers can automate.

Epilogue: The frontier is correction

The history of machine learning is often narrated as an expansion of what can be generated or predicted. The next phase may be better understood as an expansion of what can be corrected. A system becomes scientifically useful when its errors are not merely punished by a benchmark but converted into evidence about representations, assumptions, instruments and questions. The decisive capability is not permanent confidence. It is organized revisability.

Novelty enters this process as a provisional mismatch. A candidate differs from a record, distribution, class or theory. The system then asks which interpretation is warranted. Is the mismatch noise, drift, an unknown class, a new mechanism or a duplicate under another description? Each answer implies a different test. The novelty detector is therefore only the first sensor in a longer epistemic loop.

Structural limits remain. No model can create information that has no trace, choose uniquely among observationally identical worlds or certify global priority from an incomplete archive. These limits are useful because they force better questions. They motivate interventions, new sensors, scoped claims and explicit assumptions. Intelligence is partly the ability to identify when the problem must be changed.

Movable boundaries remain equally important. Models can become better calibrated under shift. Representations can become more causal and modular. Continual learning can preserve provenance. World models can expose uncertainty. Search can become more diverse. Scientific agents can learn to seek refutation and revise beliefs. Laboratories can automate controls, replication and semantic provenance. These advances do not abolish epistemology; they implement it.

The most reliable discoveries described in this survey share an architecture. A generator proposes more than one possibility. A search process explores. An evaluator provides independent resistance. An archive preserves alternatives and lineage. Experiments connect the system to the world. Institutions decide which results enter the record. Remove any one component and apparent discovery can become fluent speculation, benchmark gaming or untraceable automation.

This architecture also explains why humans are unlikely to disappear from science in one clean step. Human roles move as tools strengthen. Formal checking delegates proof verification. Autonomous laboratories delegate execution and some experiment selection. Literature systems delegate retrieval and synthesis. Humans continue to reframe questions, negotiate significance, inspect anomalies and govern consequences. Future systems may automate more of these roles, but the need for independent perspectives will remain even if their implementation changes.

The deepest novelty may not be a molecule, theorem or algorithm. It may be a new scientific institution in which machine-readable claims, executable evidence, autonomous experiments, formal verifiers and human judgment form one revisable network. Such an institution could make replication routine, negative results reusable and disagreement computationally productive. It could also centralize attention and reproduce shared blind spots. Its design is therefore a scientific and political question.

A good discovery machine should be difficult to impress and easy to correct. It should distinguish unfamiliarity from importance, confidence from evidence and execution from understanding. It should report the boundary of its claim. It should know which external test has authority and preserve the failures that led to revision. It should expand the search space without expanding the space of untraceable assertions at the same rate.

The frontier of novelty is not reached when generation becomes limitless. It is reached when criticism, verification and revision become equally scalable.

That frontier is attainable in pieces. The work now is to connect those pieces while keeping their epistemic roles distinct. Machine learning can help science discover more, but its greatest contribution may be to make the path from surprise to knowledge more explicit than it has ever been.

APPENDIX A

Appendix A. A protocol for evaluating a claim of novelty

A novelty claim should be decomposed before it is accepted or rejected. The protocol below is designed for papers, model outputs, laboratory campaigns and algorithmic discoveries. It does not produce a universal novelty score. It identifies which claim is being made, what evidence supports it and where uncertainty remains.

Table A.1. Novelty-claim audit

Audit question	Evidence that should accompany the answer
What exactly is claimed to be new?	A precise object, level of description and domain of comparison.
New relative to which record?	Named training data, databases, literature corpora, patent searches or historical scope.
What counts as equivalent?	A domain-specific semantic, functional, structural or mathematical equivalence rule.
How was correctness tested?	Formal checking, execution, simulation, measurement, replication or expert assessment.
Why is the difference significant?	Improved explanation, capability, prediction, intervention, compression or access to a new region.
Which evidence could defeat the claim?	A counterexample, prior work, failed replication, alternative mechanism or measurement.
What is the provenance?	Data, code, model versions, prompts, instruments, parameters and reviewer decisions.
What remains uncertain?	Priority, generality, mechanism, translation, measurement error and boundary conditions.

The table exists to prevent the word "novel" from carrying several unsupported conclusions at once. A candidate may pass the database comparison and fail the correctness test. It may be correct but insignificant. It may be significant and already known under another representation. Reporting these dimensions separately allows a result to be valuable without overstating it.

In practice the audit should be applied at the claim level, not only to the paper or system. A single project can contain a new dataset, a familiar method, an improved result and an unverified mechanistic interpretation. Each deserves a different status. Automated tools can assist with literature comparison, code execution and provenance, but final priority and significance judgments remain scoped and revisable.

The protocol also supports automation. A discovery agent can be required to populate each field before publication or expensive experimentation. Missing fields trigger retrieval, stronger tests or human review. The output is an evidence dossier rather than a confidence score. This is slower than unrestricted generation and much faster than correcting thousands of poorly specified claims after release.

APPENDIX B

Appendix B. A blueprint for a machine-discovery system

A general discovery architecture should be modular enough to expose failure and integrated enough to learn from it. The blueprint below is not a claim that every domain needs the same implementation. It identifies functions that should be assigned deliberately rather than hidden inside a conversational agent.

Table B.1. Functional modules of a discovery system

Module	Required responsibility
Problem and claim model	Represent the question, target, assumptions, equivalence rules and acceptable evidence.
Generator	Produce diverse candidates with lineage and declared dependencies.
Search controller	Allocate resources across performance, uncertainty, diversity and model criticism.
Evaluator ladder	Combine learned filters, execution, formal checks, simulation, experiments and review.
Experiment planner	Choose safe, discriminating observations under cost and risk constraints.
Layered memory	Preserve episodes, claims, methods, alternatives, failures and governance history.
Revision engine	Update confidence, representations and objectives while keeping versions reversible.
Provenance and governance	Log actions, enforce permissions, disclose automation and assign responsibility.

The table exists because "agent" is not a sufficient architectural description. A system can appear coherent while the same model generates, judges and summarizes every result. Explicit modules make epistemic independence testable. They also allow stronger tools to replace weaker ones without rebuilding the entire system.

The evaluator ladder should dominate deployment design. Cheap evaluators filter the broad candidate space; strong evaluators are reserved for disputed or high-value cases. Feedback should include counterexamples and failure causes, not only scalar rewards. The archive should preserve candidates across behavioural niches so that later evidence can reactivate them.

The memory system should distinguish immutable records from revisable beliefs. Raw data, instrument logs and executed code should remain fixed. Claims and confidence should change through versioned updates. Procedures should be executable and reviewed. Governance records should reveal who or what authorized each consequential action.

Safe autonomy should begin narrow: broad simulation, but approval for physical action, publication and self-modification. Expansion requires demonstrated reliability.

APPENDIX C

Appendix C. Terminology and abbreviation register

This register summarizes technical terms after they have been defined in context. It is intended for reference and editorial audit. The running text does not rely on the reader consulting this appendix to decode an abbreviation.

Table C.1. Terms and abbreviations used in the book

Term	Definition in this survey
AI — artificial intelligence	The broad field of computational systems that perform tasks associated with perception, reasoning, language, planning or action.
ML — machine learning	Methods whose behaviour is shaped substantially by data, optimization or experience.
LLM — large language model	A model trained to represent and generate language at large scale, often extended with tools and memory.
DGM — Darwin Gödel Machine	A research system that evolves and empirically evaluates modified versions of its own coding-agent scaffold.
ERA — Empirical Research Assistance	An agentic system that searches for and implements empirical scientific software against measurable evaluators.
SWE-bench — Software Engineering benchmark	A family of execution-based tasks in which coding agents attempt to resolve real issues in software repositories.
OOD — out-of-distribution	An input or process that does not arise from the declared reference distribution; always relative to a setting.
ID — in-distribution	Data treated as arising from the reference distribution used for training or evaluation.
AUROC — area under the receiver operating characteristic curve	A threshold-independent measure of how a score ranks two classes across decision thresholds.
RL — reinforcement learning	Learning a policy from interaction and evaluative feedback such as rewards.
RAG — retrieval-augmented generation	Generation conditioned on evidence retrieved from an external corpus.

Term	Definition in this survey
QD — quality-diversity	Search that preserves high-performing solutions across multiple behavioural niches.
MAP-Elites	Multi-dimensional Archive of Phenotypic Elites, a quality-diversity algorithm that stores an elite in each behavioural cell.
POET	Paired Open-Ended Trailblazer, a system that coevolves environments and the agents that solve them.
SDL — self-driving laboratory	A laboratory that connects models, planning, robotics, instruments and analysis in a closed loop.
XRD — X-ray diffraction	A measurement technique used to infer crystalline structure and phase composition.
ADMET	Absorption, distribution, metabolism, excretion and toxicity properties relevant to drug development.
CRISPR	Clustered regularly interspaced short palindromic repeats, used here in reference to gene-editing and perturbation systems.
FAIR	Findable, Accessible, Interoperable and Reusable principles for scientific data and digital objects.
IMO	International Mathematical Olympiad, a competition used in several evaluations of mathematical reasoning systems.
SMT	Satisfiability modulo theories, automated reasoning over logical formulas with background theories.
Calibration	Agreement between stated predictive confidence and observed frequency under a declared population.
Conformal prediction	A family of methods that constructs prediction sets with coverage guarantees under stated assumptions.
Identifiability	The property that the target is uniquely determined by the available evidence and assumptions.
Open-set recognition	Classification in which inputs from unknown classes may be rejected rather than forced into known classes.
Novelty detection	Detection of departures from a learned description of normal or known data.
Distribution shift	A change between the data-generating process used for training or validation and the process encountered later.

Term	Definition in this survey
World model	A learned representation used to predict states and consequences of actions in an environment.
Causal representation learning	Learning variables and mechanisms intended to support intervention and transfer rather than association alone.
Catastrophic forgetting	Loss of previously learned competence after updating on new tasks or data.
Provenance	The traceable lineage of data, code, models, tools, decisions and evidence supporting a result.
Verifier	A system that tests a candidate through formal checking, execution, simulation, measurement or review.
Open-endedness	Sustained generation of new behaviours, tasks or representations not exhausted by a fixed objective.

The table exists as a final check against hidden vocabulary. It also reveals that many abbreviations name conditional settings rather than substances. Out-of-distribution is not a property independent of a reference distribution. Calibration is not global without a population. Autonomy is layered. The definitions are therefore intentionally operational and scoped.

Terms evolve, and neighbouring research communities use them differently. Readers should prefer the explicit definition attached to a study over the familiarity of an acronym. A major editorial principle of this book is that compressed language must not compress away assumptions.

APPENDIX D

Appendix D. Methodological limits of this AI-produced survey

This survey was generated through iterative synthesis rather than through the lived expertise of a single domain specialist. That creates both advantages and risks. The system can connect literatures at scale and apply a consistent conceptual framework. It can also miss tacit context, reproduce errors in sources, overvalue accessible English-language publications and smooth over disagreements that deserve sharper treatment.

The source base was current to 24 July 2026, but the frontier changes rapidly. Several important results discussed here were recent preprints or system reports. Their inclusion is justified by the subject of the book, yet their status is stated so that novelty is not confused with confirmation. Readers should verify current versions, corrections and replications before relying on a technical claim.

The prose was audited for first-use definitions of abbreviations and technical terms. Automated scans can detect capitalized tokens, but they cannot guarantee conceptual clarity. The stronger safeguard is local explanation: what problem a term names, what assumptions it carries and what it does not imply. The diagrams were similarly redesigned as conceptual graphs and followed by interpretive prose to prevent visual metaphor from replacing argument.

The appropriate use of the book is comparative and generative. It offers a map of limits, methods and research directions that can support teaching, project design and literature review. It should invite readers to inspect the cited primary work, dispute classifications and improve the architecture. An AI-produced survey is most valuable when its provenance and limitations are explicit enough for human and machine critics to revise it.

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